

Asymptotic product structure and collar essential spectrum on $SU(2)$ instanton moduli

Vico Bonfioli
vicobonfioli@gmail.com

July 10, 2026

Abstract

On the moduli space $\mathcal{M}_k(S_r^4)$ of charge- k anti-self-dual $SU(2)$ connections on the round 4-sphere of radius r , equipped with the natural L^2 metric, we record two spectral results. First, the charge-one case: under the Habermann/Doi–Matsumoto–Matsumoto identification $\mathcal{M}_1(S_r^4) \cong \mathbb{H}_r^5$, the L^2 -Laplacian has purely continuous spectrum $[4/r^2, \infty)$, saturating McKean’s inequality. Second, the higher-charge collar: a Schwinger-parametrized closed form for the BPST scale-derivative cross-term gives the exact metric asymptotic $48\pi^2 s_1 s_2 / R^2 + O((s/R)^4)$, from which the codimension- j Uhlenbeck collar of $\mathcal{M}_k(S_r^4)$ has essential-spectrum bottom $4j/r^2$ via a Weyl quasi-mode argument on the asymptotic product of hyperbolic cusps.

1 Introduction

Let $\mathcal{M}_k(S_r^4)$ denote the moduli space of charge- k anti-self-dual $SU(2)$ connections on the round 4-sphere of radius r , equipped with the natural L^2 Hitchin-like metric inherited from the configuration space of connections (Groisser–Parker [2, 3], Habermann [4], Doi–Matsumoto–Matsumoto [5]). The geometry of these moduli spaces is asymptotically hyperbolic near the Uhlenbeck strata: at $k = 1$, the moduli space is globally a hyperbolic 5-space; at $k \geq 2$, the geometry asymptotes to a Riemannian product of bubble factors and a residual moduli factor in each Uhlenbeck collar.

We make this picture quantitative in two steps. Theorem 3.2 records the saturation of McKean’s inequality on $\mathcal{M}_1(S_r^4)$, with explicit radius dependence: $\lambda_0(\mathcal{M}_1(S_r^4)) = 4/r^2$, equal to the bottom of continuous spectrum. Lemma 4.1 then gives an exact Schwinger-parametrized closed form for the L^2 inner product of two BPST scale-derivative tangent vectors on $\mathbb{R}^4$ centered at separation R with scales s_1, s_2 :

$$\langle \partial A_1 / \partial s_1, \partial A_2 / \partial s_2 \rangle_{L^2(\mathbb{R}^4)} = 48\pi^2 s_1 s_2 \int_0^1 t(1-t) \frac{2X(t) - t(1-t)R^2}{X(t)^2} dt,$$

with leading asymptotic $48\pi^2 s_1 s_2 / R^2$ as $s_i \ll R$. The linear-in- s_i vanishing controls the cross-term in the asymptotic-product collar geometry, and Theorem 4.3 extracts the codimension- j collar essential-spectrum bottom $4j/r^2$ via a Weyl quasi-mode argument on the product of j hyperbolic cusps.

What this paper does not do. The collar essential bottom $4j/r^2$ is a statement about the spectrum localized to each Uhlenbeck stratum, not about the global $\lambda_0(\mathcal{M}_k(S_r^4))$; by domain monotonicity the global bottom is bounded above by $4/r^2$ via the shallowest collar regardless of k (Remark 4.4). Determining the global $\lambda_0(\mathcal{M}_k)$ for $k \geq 2$ is open. The body of the moduli space carries

interior curvature variation that neither McKean nor the asymptotic-product argument bounds directly.

Computational verification. The closed form (1) is verified numerically against the original 4-dimensional integral by uniform-box Monte Carlo [13].

2 Setup

Let (M, g) be a closed oriented Riemannian 4-manifold, G a compact simple Lie group, $P \rightarrow M$ a principal G -bundle. The Yang–Mills action $S_{YM}[A] = \frac{1}{2g_{YM}^2} \int_M |F_A|^2 d\text{vol}$ has linearized Hessian at any smooth ASD connection A acting in Coulomb gauge ($d_A^* a = 0$) as the restriction of the covariant Laplacian to co-closed 1-forms. The moduli space of charge- k ASD connections $\mathcal{M}_k(M) = \{A : F_A^+ = 0, c_2(P) = k\}/\mathcal{G}$ inherits an L^2 Riemannian metric from the configuration space:

$$g_{\mathcal{M}_k}(a, b) = \int_M \langle a, b \rangle d\text{vol}, \quad a, b \in T_{[A]}\mathcal{M}_k = \ker d_A^* \cap \ker d_A^+.$$

The Uhlenbeck compactification adds boundary strata at which one or more instantons bubble off, with scale parameter $s_i \rightarrow 0$.

3 The $k = 1$ case: McKean saturation on $\mathcal{M}_1(S_r^4)$

Let S_r^4 denote the round 4-sphere of radius r .

Proposition 3.1 (Habermann [4], Doi–Matsumoto–Matsumoto [5]). *The moduli space of charge-one $\text{SU}(2)$ instantons on S_r^4 with the natural L^2 metric is isometric to a hyperbolic 5-space of constant sectional curvature $-1/r^2$:*

$$\mathcal{M}_1(S_r^4) \cong \mathbb{H}_r^5 = (\text{SO}(5, 1)/\text{SO}(5), r^2 g_{\text{std}}),$$

where g_{std} is the standard hyperbolic metric of curvature -1 .

Normalization bridge. We use the unreduced L^2 norm on $\mathfrak{su}(2)$ -valued 1-forms throughout. With this choice, the per-bubble scale-derivative self-norm of a BPST instanton on $\mathbb{R}^4$ is $\|\partial A/\partial s\|_{L^2(\mathbb{R}^4)}^2 = 16\pi^2$ (used in Lemma 4.1 below), while the Habermann/DMM moduli metric of Proposition 3.1 reads $g_{\mathcal{M}_1} = (48\pi^2/\rho^2)(d\rho^2 + dx_0^2)$ in scale-position coordinates $(\rho, x_0) \in \mathbb{R}^+ \times \mathbb{R}^4 \subset \mathbb{H}^5$. The factor of 3 between $16\pi^2$ and $48\pi^2$ reflects the η -tensor contraction $\sum_{a,\mu} \eta_{\mu\nu}^a \eta_{\mu\rho}^a = 3\delta_{\nu\rho}$ (equivalently, the trace over $\mathfrak{su}(2)$); curvature $-1/r^2$ then follows from the standard $r^2 g_{\text{std}}$ rescaling. The two conventions are used consistently within their respective sections.

Theorem 3.2 (Spectral bottom on $\mathcal{M}_1(S_r^4)$). *The spectrum of the natural L^2 -Laplacian on $\mathcal{M}_1(S_r^4) \cong \mathbb{H}_r^5$ is purely continuous, equal to $[4/r^2, \infty)$, with no L^2 eigenfunctions. The spectral bottom is*

$$\lambda_0(\mathcal{M}_1(S_r^4)) = \frac{4}{r^2} = \frac{(n-1)^2}{4} \cdot \frac{1}{r^2}$$

with $n = 5$, saturating McKean’s inequality [6].

Proof. On unit hyperbolic space $\mathbb{H}^5$ in upper-half-space coordinates $\{(x_1, \dots, x_4, y) : y > 0\}$ with metric $g_{\text{std}} = (dx^2 + dy^2)/y^2$, the volume density is $\sqrt{\det g} = y^{-5}$ and the Laplace–Beltrami operator on functions $f(y)$ depending only on the radial coordinate reduces, after the standard computation $\Delta_g f = (\det g)^{-1/2} \partial_y ((\det g)^{1/2} g^{yy} \partial_y f)$, to

$$\Delta_{\mathbb{H}^5} f = \frac{1}{r^2} [y^2 f''(y) - 3y f'(y)] \quad \text{on radial } f \text{ on } \mathbb{H}_r^5.$$

The radial generalized eigenfunctions $\phi_s(y) = y^s$ then satisfy $-\Delta_{\mathbb{H}^5} \phi_s = s(4-s)\phi_s$, maximized at $s = 2$ with eigenvalue $\mu = 4$. The volume measure $d\text{vol} = y^{-5} dx dy$ gives $\int |\phi_s|^2 d\text{vol} = \int y^{2s-5} dx dy$, divergent for all s , so the bottom $\lambda_0(\mathbb{H}^5) = 4$ lies at the bottom of continuous spectrum. Rescaling the metric by r^2 rescales the Laplacian by r^{-2} , hence the spectrum by r^{-2} : $\lambda_0(\mathbb{H}_r^5) = 4/r^2$. McKean’s inequality [6] on an n -manifold with sectional curvature $K \leq -\kappa^2$ gives $\lambda_0 \geq (n-1)^2 \kappa^2 / 4$, equal to $4/r^2$ here; equality holds on constant-curvature spaces. $\square$

Remark 3.3. The dimensional structure is explicit: λ_0 has dimensions $[\text{length}]^{-2}$, scaling as $1/r^2$ with the radius of the underlying S^4 . The dimensionless ratio is $\lambda_0 \cdot r^2 = 4$.

4 Higher charge: collar essential spectrum via the Schwinger cross-term lemma

For $k \geq 2$ the global L^2 metric on $\mathcal{M}_k(S_r^4)$ is not known in closed form. The body of $\mathcal{M}_k$ is not globally of constant curvature, so McKean’s theorem does not directly apply globally. In each Uhlenbeck collar, however, the geometry asymptotes to a Riemannian product of bubble factors (each modelled by the cusp end of $\mathbb{H}_r^5$) and a residual background factor. The asymptotic product structure was identified qualitatively by Taubes [7] and developed in Donaldson–Kronheimer [12] §7.3; Lemma 4.1 below makes it quantitative for the L^2 -metric scale-derivative cross-term.

4.1 The Schwinger cross-term lemma

Lemma 4.1 (Cross-term decoupling in the Uhlenbeck collar; exact Schwinger closed form). *Let A_1 and A_2 be two BPST $\text{SU}(2)$ instantons on $\mathbb{R}^4$ centered at distinct points x_1, x_2 with scales s_1, s_2 and separation $R = |x_1 - x_2|$. Then the L^2 inner product of their scale-derivative tangent vectors admits the exact closed form*

$$\langle \partial A_1 / \partial s_1, \partial A_2 / \partial s_2 \rangle_{L^2(\mathbb{R}^4)} = 48\pi^2 s_1 s_2 \int_0^1 t(1-t) \frac{2X(t) - t(1-t)R^2}{X(t)^2} dt, \quad (1)$$

where

$$X(t) := t(1-t)R^2 + (1-t)s_1^2 + ts_2^2. \quad (2)$$

The integral evaluates in elementary closed form: with $\Sigma := R^2 + s_1^2 + s_2^2$ and $D := \Sigma^2 - 4s_1^2 s_2^2 = (R^2 + (s_1 + s_2)^2)(R^2 + (s_1 - s_2)^2)$,

$$\langle \partial A_1 / \partial s_1, \partial A_2 / \partial s_2 \rangle_{L^2(\mathbb{R}^4)} = 48\pi^2 s_1 s_2 \left[\frac{\Sigma}{D} - \frac{4s_1^2 s_2^2}{D^{3/2}} \text{arccosh}\left(\frac{\Sigma}{2s_1 s_2}\right) \right]. \quad (3)$$

Moreover (3) is intrinsic to the hyperbolic geometry of Proposition 3.1: $\Sigma/(2s_1 s_2) = \cosh d$, where d is the hyperbolic distance between (x_1, s_1) and (x_2, s_2) in the upper-half-space model of $\mathbb{H}^5$ (unit curvature), so

$$\langle \partial A_1 / \partial s_1, \partial A_2 / \partial s_2 \rangle_{L^2(\mathbb{R}^4)} = 24\pi^2 \frac{\cosh d \sinh d - d}{\sinh^3 d} =: 24\pi^2 G(d), \quad (4)$$

a function of the hyperbolic distance alone. G is smooth and positive on $(0, \infty)$, with $G(d) \rightarrow 2/3$ as $d \rightarrow 0^+$: the cross-term extends continuously to the coincidence limit with value $16\pi^2$, matching the diagonal norm $\|\partial A_1/\partial s_1\|_{L^2}^2 = 16\pi^2$ below. As $d \rightarrow \infty$, $G(d) = 2e^{-d} + O(de^{-3d})$, which recovers the leading term of (5) via $e^{-d} = (1 + o(1)) s_1 s_2 / R^2$ in the regime $s_1, s_2 \ll R$. In the asymptotic regime $s_1, s_2 \ll R$, the leading behaviour with explicit next-order correction is

$$\langle \partial A_1 / \partial s_1, \partial A_2 / \partial s_2 \rangle_{L^2(\mathbb{R}^4)} = \frac{48\pi^2 s_1 s_2}{R^2} \left[1 - \frac{s_1^2 + s_2^2}{R^2} \right] + O\left(\frac{s_1 s_2 (s_1^2 + s_2^2)^2 |\log(\max(s_1, s_2)/R)|}{R^6} \right). \quad (5)$$

The next-order $-(s_1^2 + s_2^2)/R^2$ coefficient is exactly -1 . The remainder carries a $|\log(\max(s_i)/R)|$ factor at order $(s/R)^4$ via the symmetric $u^2 v^2$ cross-term in the dimensionless expansion (with $u = s_1/R$, $v = s_2/R$). The on-axis slice is rational: with $F(u, v) := I/(48\pi^2 s_1 s_2 / R^2)$ for $s_1 s_2 > 0$, the limit $\lim_{v \rightarrow 0} F(u, v) = 1/(1 + u^2)$ holds, with no logarithm in the $v \equiv 0$ slice. In particular, fixing (R, s_2) and letting $s_1 \rightarrow 0$, the cross-term vanishes linearly in s_1 , and the corresponding L^2 -metric component (after normalization by the diagonal norm $\|\partial A_1/\partial s_1\|_{L^2(\mathbb{R}^4)}^2 = 16\pi^2$, which is s_1 -independent) also vanishes linearly in s_1 . Normalization note: we use the unreduced L^2 norm on $\mathfrak{su}(2)$ -valued 1-forms (so the per-bubble L^2 self-norm is $16\pi^2$); the Habermann/Groisser-Parker normalization of the hyperbolic moduli metric (giving the curvature $-1/r^2$ via $g_{\mathcal{M}_1} = (48\pi^2/\rho^2)(d\rho^2 + dx_0^2)$) differs by a fixed factor and is used consistently in Theorem 3.2.

Proof. After the η -tensor contraction identity $\sum_{a,\mu} \eta_{\mu\nu}^a \eta_{\mu\rho}^a = 3\delta_{\nu\rho}$ for the self-dual 't Hooft symbols, the inner product reduces to

$$\langle \partial A_1 / \partial s_1, \partial A_2 / \partial s_2 \rangle_{L^2} = 48s_1 s_2 \int_{\mathbb{R}^4} \frac{x \cdot (x - Re_1)}{(|x|^2 + s_1^2)^2 ((x - Re_1)^2 + s_2^2)} d^4 x.$$

Use the Schwinger parametrization $A^{-2} = \int_0^\infty \alpha e^{-\alpha A} d\alpha$ for each squared denominator. Set $\xi := x - \xi_*$ with the shift $\xi_* := (\alpha_2 R / (\alpha_1 + \alpha_2)) e_1$ (we rename the integration variable to ξ to avoid clash with the upper-half-space coordinate y used later in the spectral analysis). Completing the square in x gives the integrand as $\exp(-(\alpha_1 + \alpha_2)|\xi|^2 - \alpha_1 \alpha_2 R^2 / (\alpha_1 + \alpha_2) - \alpha_1 s_1^2 - \alpha_2 s_2^2)$. Evaluate the Gaussian ξ -integrals using $\int_{\mathbb{R}^4} e^{-\sigma|\xi|^2} d^4 \xi = \pi^2/\sigma^2$, $\int |\xi|^2 e^{-\sigma|\xi|^2} d^4 \xi = 2\pi^2/\sigma^3$, $\int \xi_1 e^{-\sigma|\xi|^2} d^4 \xi = 0$. Repartition $(\alpha_1, \alpha_2) \rightarrow (\sigma, t)$ with $\sigma = \alpha_1 + \alpha_2$, $t = \alpha_2/\sigma$ (Jacobian σ), then integrate σ from 0 to ∞ using $\int_0^\infty e^{-\sigma X} d\sigma = 1/X$ and $\int_0^\infty \sigma e^{-\sigma X} d\sigma = 1/X^2$. The result is exactly (1).

For the elementary form (3), write $a = s_1^2$, $b = s_2^2$, $c = R^2$ and $L(t) := a(1-t) + bt$, so that $X = ct(1-t) + L$ and the integrand of (1) collapses pointwise:

$$t(1-t) \frac{2X - t(1-t)c}{X^2} = \frac{(X-L)(X+L)}{cX^2} = \frac{1}{c} \left(1 - \frac{L^2}{X^2} \right).$$

The quadratic $X(t) = -ct^2 + (c+b-a)t + a$ is positive on $[0, 1]$ (both roots lie outside the interval, since $X(0) = a > 0$, $X(1) = b > 0$ and the leading coefficient is negative) with $(c+b-a)^2 + 4ac = \Sigma^2 - 4ab = D$. The standard antiderivatives give

$$\int_0^1 \frac{dt}{X} = \frac{2}{\sqrt{D}} \operatorname{arccosh}\left(\frac{\Sigma}{2\sqrt{ab}}\right), \quad \int_0^1 \frac{X' dt}{X^2} = \frac{1}{a} - \frac{1}{b}, \quad \int_0^1 \frac{dt}{X^2} = \frac{(a-b)^2 + c(a+b)}{abD} + \frac{2c}{D} \int_0^1 \frac{dt}{X}.$$

Decomposing $L^2 = AX + BX' + C$ with rational coefficients ($A = -(b-a)^2/c$, and B, C determined by matching the linear and constant terms), assembling the three integrals, and using the polynomial identity

$$a^3 - a^2b + 3a^2c - ab^2 + 2abc + 3ac^2 + b^3 + 3b^2c + 3bc^2 + c^3 = \Sigma D,$$

one obtains (3). The substitution $\Sigma = 2s_1s_2 \cosh d$, $\sqrt{D} = 2s_1s_2 \sinh d$ then gives (4); that d is the $\mathbb{H}^5$ distance is the upper-half-space formula $\cosh d = 1 + (|x_1 - x_2|^2 + (s_1 - s_2)^2)/(2s_1s_2) = \Sigma/(2s_1s_2)$. The limits of G follow from $\cosh d \sinh d - d = \frac{2}{3}d^3 + O(d^5)$ and $\sinh^3 d = d^3 + O(d^5)$ at $d \rightarrow 0$, and from $\cosh d \sinh d - d = \frac{1}{4}e^{2d}(1 - 4de^{-2d} + O(e^{-4d}))$, $\sinh^3 d = \frac{1}{8}e^{3d}(1 + O(e^{-2d}))$ at $d \rightarrow \infty$. The identities and (3)–(4) are verified symbolically and to 50 significant digits against direct quadrature of (1), and to machine precision against the 4-D grid of [14], in [16].

The asymptotic (5) follows by expanding $1/X(t)$ in powers of $(s_i/R)^2$ in the integrand and integrating term by term against $t(1-t)$. The leading 1 and the explicit $-(s_1^2 + s_2^2)/R^2$ correction follow from elementary $\int_0^1 t(1-t) dt = 1/6$ identities; the $\log(s/R)$ at the next order arises from the endpoint behaviour of the u^2v^2 contribution where $X(t) \sim ts_2^2$ as $t \rightarrow 0$ and $X(t) \sim (1-t)s_1^2$ as $t \rightarrow 1$, producing logarithmic divergence after the polynomial $t(1-t)$ damping is exhausted. Numerical verification of (1) against the original 4-dimensional integral has been performed by adaptive Gauss–Kronrod cubature (after a reduction to two dimensions exploiting the $O(3)$ symmetry around the axis through x_1, x_2): the closed form agrees with the 4D integral to ≈ 2.5 ulp ($\approx 5.5 \times 10^{-16}$ relative error) across a $5 \times 5 \times 4 + 6$ grid of (s_1, s_2, R) values spanning four orders of magnitude in scale ratio [14]. The on-axis closed form $F(u, 0) = 1/(1 + u^2)$ and the symmetric next-order coefficients $c_4 = d_4 = 8$ in $F(\epsilon, \epsilon) = 1 - 2\epsilon^2 + 8(1 + \log \epsilon)\epsilon^4 + O(\epsilon^6 \log \epsilon)$ are confirmed symbolically in [15]. $\square$

4.2 Off-diagonal cross-blocks: position derivatives and the gauge correction

To compare $g_{U_\epsilon^{(j)}}$ to a Riemannian product we must control *all* bubble-bubble matrix elements of the L^2 metric, not only the scale-scale entries of Lemma 4.1. The position-derivative tangent vectors require an additional ingredient: the *horizontal* (Coulomb-gauge) projection. The bare partial derivative $\partial A/\partial x_0^\rho$ of a BPST instanton on $\mathbb{R}^4$ contains a long-range $|x - x_0|^{-2}$ tail (the rotation gauge field) which is *not* square-integrable individually; only the horizontal projection onto $\ker d_A^*$ is. We record the closed forms and decay rates of the relevant bubble-bubble inner products in the bare and gauge-fixed normalizations.

Lemma 4.2 (Off-diagonal bubble cross-block bounds). *Let A_1, A_2 be two BPST $SU(2)$ instantons on $\mathbb{R}^4$ centered at x_1, x_2 with separation $R = |x_1 - x_2|$ and scales s_1, s_2 , and write $v_s^{(i)} = \partial A_i/\partial s_i$ and $v_{x,\rho}^{(i)} = \partial A_i/\partial x_i^\rho$ for the naive tangent vectors. Let $\hat{v}_{x,\rho}^{(i)}$ denote the L^2 -orthogonal projection onto $\ker d_{A_i}^*$ (Coulomb-horizontal projection).*

- (i) (Scale-scale; Lemma 4.1.) $\langle v_s^{(1)}, v_s^{(2)} \rangle_{L^2} = 48\pi^2 s_1 s_2 / R^2 + O(s_1 s_2 (s/R)^4 \log(s/R))$.
- (ii) (Scale-position, bare.) *Only the axial component $\rho = 1$ (along $x_2 - x_1$) is nonzero by $O(3)$ -symmetry; the orthogonal components vanish identically. The axial component admits an exact Schwinger representation (recorded in [23]; involving $1/X(t)^2$ -type denominators from the squared $1/D_i^2$ propagators after σ -integration). Its leading asymptotic in $s_1, s_2 \ll R$ is*

$$\langle v_s^{(1)}, v_{x,1}^{(2)} \rangle_{L^2(\mathbb{R}^4)} = \frac{72\pi^2 s_1}{R} - 4\pi^2 s_1 R + O(s_1 s_2^2 / R). \quad (6)$$

The linear-in- R piece is the long-range tail of $v_{x,1}^{(2)}$ before gauge fixing. The exact closed form is not used in the sequel; only the gauge-fixed bound (iv) below enters the proofs of Theorems 4.3–4.8.

- (iii) (Position-position, bare.) *The naive inner product $\langle v_{x,\rho}^{(1)}, v_{x,\sigma}^{(2)} \rangle_{L^2(\mathbb{R}^4)}$ is logarithmically infrared-divergent for $s_1, s_2 > 0$: the integrand contains a $\int (x \cdot (x - Re_1))^2 / D_1^2 D_2^2 d^4x$ piece with $|x|^{-4}$ tail at infinity. The horizontal projections $\hat{v}_{x,\rho}^{(i)}$, however, are square-integrable.*

(iv) (Gauge-fixed off-diagonal bound.) *In the moduli L^2 -metric, the gauge-fixed bubble-bubble cross-block has matrix entries bounded by*

$$|\langle \hat{v}_\alpha^{(1)}, \hat{v}_\beta^{(2)} \rangle_{L^2}| \leq \frac{C s_1 s_2}{R^2} \|\hat{v}_\alpha^{(1)}\|_{L^2} \cdot \|\hat{v}_\beta^{(2)}\|_{L^2} \cdot (1 + |\log(\min(s_1, s_2)/R)|), \quad (7)$$

uniformly in $s_1, s_2 \leq cR$ for some absolute $C, c > 0$, with $\alpha, \beta \in \{s, x_1, x_2, x_3, x_4\}$ indexing the five bubble tangent directions and $\hat{v}_s = v_s$.

(v) (Bubble-background cross-block.) *Let a be any tangent vector to the residual moduli factor $\mathcal{M}_{k-j}$, supported (in S^4 coordinates) at distance $\geq R$ from a bubble center x_i with bubble scale s_i . Then $|\langle a, \hat{v}_\alpha^{(i)} \rangle_{L^2}| \leq C (s_i^2/R) \|a\|_{L^2} \cdot \|\hat{v}_\alpha^{(i)}\|_{L^2}$.*

Proof. Part (i) is Lemma 4.1. For (ii), the BPST derivative computation gives

$$v_s^{(1)} = -4s_1 \eta_{\mu\nu}^a x^\nu / D_1^2, \quad v_{x,\rho}^{(2)} = -2\eta_{\mu\rho}^a / D_2 + 4\eta_{\mu\nu}^a (x - Re_1)^\nu (x - Re_1)^\rho / D_2^2,$$

with $D_i = |x - x_i|^2 + s_i^2$. The 't Hooft contraction $\sum_{a,\mu} \eta_{\mu\nu}^a \eta_{\mu\rho}^a = 3\delta_{\nu\rho}$ collapses the spin index and pins the surviving direction to $\rho = 1$ (the line $x_1 x_2$); the transverse $\rho \in \{2, 3, 4\}$ integrands are odd under reflection $x_\rho \rightarrow -x_\rho$ and vanish. The two resulting scalar integrals

$$24s_1 \int_{\mathbb{R}^4} \frac{x_1}{D_1^2 D_2} d^4x, \quad -48s_1 \int_{\mathbb{R}^4} \frac{(x \cdot (x - Re_1))(x_1 - R)}{D_1^2 D_2^2} d^4x$$

are evaluated by the Schwinger parametrization of Lemma 4.1 (each squared denominator $1/D^2 = \int_0^\infty \alpha e^{-\alpha D} d\alpha$, each linear denominator $1/D = \int_0^\infty e^{-\alpha D} d\alpha$). Completing the square in x with $\sigma = \alpha_1 + \alpha_2$, $t = \alpha_2/\sigma$ shifts the Gaussian center to $\xi_* = (\alpha_2 R/\sigma)e_1$, and the standard Gaussian moments $\int e^{-\sigma|\xi|^2} = \pi^2/\sigma^2$, $\int |\xi|^2 e^{-\sigma|\xi|^2} = 2\pi^2/\sigma^3$, $\int \xi_1^2 e^{-\sigma|\xi|^2} = \pi^2/(2\sigma^3)$ produce an exact closed form involving denominators $X(t)^2$ (recorded in [23]). The leading asymptotic (6) follows by substituting $X(t) \rightarrow t(1-t)R^2$ and using $\int_0^1 t(1-t) dt = 1/6$, $\int_0^1 (1-t) dt = 1/2$, $\int_0^1 t(1-t)^2 dt = 1/12$. Numerical verification [23] confirms (6) to $< 10^{-3}$ relative error across (R, s_1, s_2) ranging over four orders of magnitude.

For (iii), $|v_{x,\rho}^{(i)}|^2 \sim 4|\eta_{\mu\rho}^a|^2/D_i^2 \sim 12/|x|^4$ at large $|x|$, so $\int |v_{x,\rho}|^2 d^4x$ diverges logarithmically at infinity. The trace integrand of $\langle v_{x,\rho}^{(1)}, v_{x,\sigma}^{(2)} \rangle$ inherits a $|x|^{-4}$ infrared tail directly.

For (iv), the horizontal projection $\hat{v} = v - d_A G_A d_A^* v$, where G_A is the Green's function for the Laplacian on $\mathfrak{g}$ -valued functions in the BPST background, kills the long-range $|x - x_0|^{-2}$ piece of $v_{x,\rho}$, leaving $|\hat{v}_{x,\rho}(x)| = O(s^2/|x - x_0|^3)$ at infinity (the field-strength rate). The cross-block bound (7) then follows from a weighted Sobolev pairing argument in the Taubes/Donaldson–Kronheimer framework, with the log factor entering through a borderline radial integral over the inter-bubble annulus $\min(s_1, s_2) \lesssim |x - x_i| \lesssim R$. The full proof, including the scale-uniformity of the weighted Coulomb estimate and the tracking of the log factor, is given in Appendix A.

For (v): we apply the same horizontal-projection localization (Appendix A, §A.5). The tangent vector $\hat{v}_\alpha^{(i)}$ has L^2 mass concentrated in a ball of radius $O(s_i)$; pairing with a residual tangent vector a supported at distance $\geq R$ from x_i gives the stated s_i^2/R rate via the four-dimensional Sobolev embedding $H^2 \hookrightarrow L^\infty$ together with the Coulomb-tail estimate of step (iv). $\square$

4.3 Collar essential-spectrum bottom

Theorem 4.3 (Collar-localized essential-spectrum bottom on $\mathcal{M}_k(S_r^4)$). *Let $\mathcal{M}_k(S_r^4)$ be the k -instanton moduli for $SU(2)$ on S_r^4 , equipped with the L^2 metric, and let $U_\epsilon^{(j)} \subset \mathcal{M}_k$ denote the*

tubular neighborhood of the Uhlenbeck stratum at which exactly j instantons have scale parameter below ϵ (the codimension- j collar). Then the collar-localized essential-spectrum bottom satisfies

$$\lim_{\epsilon \rightarrow 0} \lambda_0^{\text{ess}}(U_\epsilon^{(j)}) = \frac{4j}{r^2} \quad \text{for each } 1 \leq j \leq k.$$

In particular, the shallowest collar (one bubbling instanton) contributes essential spectrum at $4/r^2$, and the deepest stratum (all k instantons simultaneously bubbling) contributes essential spectrum at $4k/r^2$.

Remark 4.4 (Domain monotonicity bounds the global bottom). By domain monotonicity, $\lambda_0(\mathcal{M}_k) \leq \lambda_0^{\text{ess}}(U_\epsilon^{(1)}) \rightarrow 4/r^2$, so $\lambda_0(\mathcal{M}_k(S_r^4)) \leq 4/r^2$ for every $k \geq 1$. Theorem 4.3 identifies the essential-spectrum contributions of each Uhlenbeck stratum but does *not* settle whether $\lambda_0(\mathcal{M}_k) = 4/r^2$ for all k , nor whether interior L^2 eigenvalues below this threshold exist. The bound $4k/r^2$ at the deepest stratum is the essential-spectrum bottom of a deep slice, not the bottom of the full Laplacian; the latter is bounded above by $4/r^2$ regardless. The two are distinct quantities and should not be conflated.

Proof of Theorem 4.3. By Lemma 4.2 (clauses (i), (iv), (v)) the L^2 metric on $U_\epsilon^{(j)}$ admits an off-diagonal cross-block bound in the gauge-fixed (horizontal) basis of bubble tangent directions: for each pair of bubble-tangent matrix entries (scale-scale, scale-position, position-position) and each bubble-background matrix entry, the absolute value is bounded by a constant multiple of $\epsilon |\log \epsilon|$ times the geometric mean of the diagonal entries on either side, where $\epsilon = \max_i s_i$. Consequently, in any orthonormal basis adapted to the diagonal block structure (bubble i and background $\mathcal{M}_{k-j}$), the metric tensor satisfies the operator-norm comparison

$$(1 - C\epsilon |\log \epsilon|) \left(g_{\mathcal{M}_{k-j}(S_r^4)} \oplus \bigoplus_{i=1}^j g_{\text{bubble},i} \right) \leq g_{U_\epsilon^{(j)}} \leq (1 + C\epsilon |\log \epsilon|) \left(g_{\mathcal{M}_{k-j}(S_r^4)} \oplus \bigoplus_{i=1}^j g_{\text{bubble},i} \right), \quad (8)$$

for some constant C depending only on the geometry of S_r^4 (in particular on the lower bound for the separation R in the collar, which can be taken bounded away from zero throughout $U_\epsilon^{(j)}$ for ϵ small enough). By Proposition 3.1, each individual bubble factor is, in the limit $s_i \rightarrow 0$, isometric to the cusp end (horosphere bundle near infinity) of $\mathbb{H}_r^5$.

In fact (8) extends to a C^2 comparison of the perturbation tensor itself. Writing $h := g_{U_\epsilon^{(j)}} - g_{\text{prod}}$ and denoting by ∇^{prod} the Levi-Civita connection of the product metric, for ϵ small enough one has, uniformly on the cusp ends of $U_\epsilon^{(j)}$:

$$|h|_{g_{\text{prod}}} \leq C\epsilon |\log \epsilon|, \quad |\nabla^{\text{prod}} h|_{g_{\text{prod}}} \leq C \frac{\epsilon |\log \epsilon|}{r_{\text{cusp}}}, \quad |(\nabla^{\text{prod}})^2 h|_{g_{\text{prod}}} \leq C \frac{\epsilon |\log \epsilon|^2}{r_{\text{cusp}}^2}, \quad (9)$$

where r_{cusp} is the (inverse-length) cusp scale on the relevant bubble factor (in upper-half-space coordinates, $r_{\text{cusp}}^{-1} \sim y_i \partial_{y_i}$ acting as a C^0 -bounded unit). This C^2 statement is the $n = 0, 1, 2$ specialization of Lemma A.6: each application of ∇^{prod} in cusp-adapted coordinates is equivalent (up to bounded coefficients of the $\mathbb{H}_r^5$ connection symbols) to a single application of the cusp-dilation derivative $y_i \partial_{y_i}$, which by Lemma A.6 costs at most one extra logarithmic factor per derivative and no inverse power of the bubble scale.

Upper bound on the essential bottom: Weyl quasi-modes. The cusps of $\mathbb{H}_r^5$ have purely continuous spectrum with no L^2 ground state. In upper-half-space coordinates $(\xi_1^{(i)}, \dots, \xi_4^{(i)}, y_i)$ for the i -th cusp ($i = 1, \dots, j$), with $y_i > 0$, the radial generalized eigenfunctions of the Laplace–Beltrami

operator on each cusp are $\phi_s^{(i)}(y_i) = y_i^s$, with eigenvalue $r^{-2}s(4-s)$ maximized at $s = 2$ with value $4/r^2$. Choose smooth cutoffs $\chi_n \in C_c^\infty(\mathbb{R})$ supported in $[-n, n]$ with $\chi_n \equiv 1$ on $[-n+1, n-1]$ and $\|\chi_n'\|_\infty + \|\chi_n''\|_\infty \leq C$. Define

$$\Phi_n^{(j)}(y_1, \dots, y_j) = \prod_{i=1}^j y_i^2 \chi_n(\log y_i).$$

Direct calculation of $-\Delta \Phi_n^{(j)}$ on the *exact* product cusp gives $(-\Delta_{\text{prod}} + 4j/r^2)\Phi_n^{(j)} = O(\chi_n') + O(\chi_n'')$, and the routine estimate $\|\chi_n\|_{L^2} \sim \sqrt{n}$, $\|\chi_n'\|_{L^2} = O(1)$ yields

$$\frac{\|(-\Delta_{\text{prod}} - 4j/r^2)\Phi_n^{(j)}\|_{L^2}}{\|\Phi_n^{(j)}\|_{L^2}} \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

We transport this quasi-mode into $U_\epsilon^{(j)}$ using (8). The Laplace–Beltrami operator $\Delta_{U_\epsilon^{(j)}}$ on the collar differs from Δ_{prod} by an operator of the form $L_\epsilon = (g_{U_\epsilon}^{-1} - g_{\text{prod}}^{-1})\partial\bar{\partial} + \text{lower-order}$, with $\|g_{U_\epsilon}^{-1} - g_{\text{prod}}^{-1}\|_{\text{op}} \leq C'\epsilon|\log \epsilon|$ by (8). Hence

$$\frac{\|(-\Delta_{U_\epsilon^{(j)}} - 4j/r^2)\Phi_n^{(j)}\|_{L^2}}{\|\Phi_n^{(j)}\|_{L^2}} \leq \frac{\|(-\Delta_{\text{prod}} - 4j/r^2)\Phi_n^{(j)}\|_{L^2}}{\|\Phi_n^{(j)}\|_{L^2}} + C'\epsilon|\log \epsilon| \cdot \frac{\|\nabla^2 \Phi_n^{(j)}\|_{L^2}}{\|\Phi_n^{(j)}\|_{L^2}}.$$

The second term involves $\|\nabla^2 \Phi_n^{(j)}\|_{L^2}^2 / \|\Phi_n^{(j)}\|_{L^2}^2 = O(1)$ as $n \rightarrow \infty$. Explicitly, the hyperbolic Laplacian on the cusp acts on $y_i^2 = e^{2\log y_i}$ by an order-one operator in the logarithmic coordinate $u_i = \log y_i$ (since $y_i \partial_{y_i} = \partial_{u_i}$ and the metric is $dx_i^2/y_i^2 + du_i^2$ along the cusp), so $\|\nabla^2(y_i^2 \chi_n(u_i))\|_{L^2}^2 = O(\|\chi_n\|_{L^2}^2 + \|\chi_n'\|_{L^2}^2 + \|\chi_n''\|_{L^2}^2) = O(n+1)$, while $\|y_i^2 \chi_n(u_i)\|_{L^2}^2 = O(n)$ (the $y_i^4 \cdot y_i^{-5}$ factor integrates against $du_i dx_i$ to give $O(\|\chi_n\|_{L^2}^2)$). The ratio is therefore $O(1)$, with the χ_n', χ_n'' contributions producing the $o(1)$ part of the first term in the displayed bound. Choosing $n = n(\epsilon)$ with $n(\epsilon) \rightarrow \infty$ slowly enough that the first term is $o(1)$ while $\epsilon|\log \epsilon| \rightarrow 0$ ensures both terms vanish. Hence $4j/r^2 \in \sigma_{\text{ess}}(-\Delta_{U_\epsilon^{(j)}})$ for ϵ small enough, giving the upper bound $\liminf_\epsilon \lambda_0^{\text{ess}}(U_\epsilon^{(j)}) \leq 4j/r^2$.

Lower bound: form comparison and min-max. For the matching lower bound, the operator inequality (8) on the metric tensor implies the comparison of Laplace–Beltrami operators as quadratic forms on $C_c^\infty(U_\epsilon^{(j)})$:

$$(1 - C\epsilon|\log \epsilon|)\langle -\Delta_{\text{prod}} \phi, \phi \rangle \leq \langle -\Delta_{U_\epsilon^{(j)}} \phi, \phi \rangle \leq (1 + C\epsilon|\log \epsilon|)\langle -\Delta_{\text{prod}} \phi, \phi \rangle.$$

By the min-max principle, $\lambda_0^{\text{ess}}(-\Delta_{U_\epsilon^{(j)}}) \geq (1 - C\epsilon|\log \epsilon|)\lambda_0^{\text{ess}}(-\Delta_{\text{prod}})$. The product Laplacian decomposes as $-\Delta_{\text{prod}} = -\Delta_{\mathcal{M}_{k-j}} \otimes 1 + 1 \otimes (-\Delta_{\text{bubbles}})$, and the spectrum of a tensor sum is the Minkowski sum of the constituent spectra; thus $\lambda_0^{\text{ess}}(-\Delta_{\text{prod}}) \geq \lambda_0(\mathcal{M}_{k-j}) + \sum_{i=1}^j \lambda_0(\mathbb{H}_r^5\text{-cusp}) \geq 0 + 4j/r^2$, using $\lambda_0 \geq 0$ on the (possibly compact) residual factor and the McKean saturation $\lambda_0(\mathbb{H}_r^5) = 4/r^2$ from Theorem 3.2. Taking $\epsilon \rightarrow 0$ gives $\liminf_\epsilon \lambda_0^{\text{ess}}(U_\epsilon^{(j)}) \geq 4j/r^2$. The two bounds match. $\square$

Corollary 4.5 (Quantitative two-sided rate). *There exist $C = C(j, r) > 0$ and $\epsilon_0 > 0$ such that for all $0 < \epsilon < \epsilon_0$,*

$$\left| \lambda_0^{\text{ess}}(U_\epsilon^{(j)}) - \frac{4j}{r^2} \right| \leq C \frac{\epsilon |\log \epsilon|}{r^2}.$$

Proof. The min-max step of the preceding proof gives $\lambda_0^{\text{ess}}(U_\epsilon^{(j)}) \geq (1 - C\epsilon|\log \epsilon|)4j/r^2$. Conversely, the transported quasi-modes $\Phi_n^{(j)}$ form, for $n \rightarrow \infty$ at fixed ϵ , a Weyl-type sequence whose normalized residual $\|(-\Delta_{U_\epsilon^{(j)}} - 4j/r^2)\Phi_n^{(j)}\|_{L^2}/\|\Phi_n^{(j)}\|_{L^2}$ is bounded, in the limit, by $C'\epsilon|\log \epsilon|/r^2$; hence $\text{dist}(4j/r^2, \sigma_{\text{ess}}(-\Delta_{U_\epsilon^{(j)}})) \leq C'\epsilon|\log \epsilon|/r^2$, and since the min-max bound excludes essential spectrum below $(1 - C\epsilon|\log \epsilon|)4j/r^2$, the essential-spectrum point so produced bounds λ_0^{ess} from above by $4j/r^2 + C'\epsilon|\log \epsilon|/r^2$. The two bounds combine into the display. $\square$

Remark 4.6 (On the strength of the metric-comparison rate). The remainder rate in (8) is $O(\epsilon|\log \epsilon|)$ rather than $O(\epsilon)$, owing to the logarithmic factor in the gauge-fixed position-position cross-block bound (7); this is consistent with the logarithm appearing in the symmetric next-order remainder of Lemma 4.1 (cf. [15]). The $O(\epsilon|\log \epsilon|) \rightarrow 0$ rate is amply sufficient for Theorem 4.3. The strengthened C^2 form (9), together with Lemma A.6, is what makes the Mourre/Sahbani machinery of Theorem 4.8 applicable: an operator-norm (quadratic-form) bound alone would suffice for $\inf \sigma_{\text{ess}}$ but would not control the iterated commutators required for the $C^{1,1}$ -regularity hypothesis of [17].

Remark 4.7. The body of $\mathcal{M}_k$ for $k \geq 2$ has curvature variation in the interior (away from collars), where neither McKean nor the asymptotic-product argument directly bounds the local spectrum. Theorem 4.3 is specifically a statement about the Uhlenbeck collar regions, where the geometry is asymptotically rigid.

4.4 Spectral type on the Uhlenbeck collar

The metric-comparison estimate (8) is sufficient to identify the bottom of the essential spectrum; with the iterated-commutator bounds of Appendix A, §A.7, it also identifies the *spectral type* above the threshold via Mourre's commutator method. We use the formulation due to Sahbani [17], which improves the classical Amrein–Boutet de Monvel–Georgescu framework [20] to the optimal $C^{1,1}$ -regularity class for the conjugate operator.

Theorem 4.8 (Spectral type on Uhlenbeck collars). *Let $U_\epsilon^{(j)} \subset \mathcal{M}_k(S_r^4)$ be the codimension- j Uhlenbeck collar as in Theorem 4.3. For every $\delta > 0$, there exists $\epsilon_0 = \epsilon_0(\delta, j, r) > 0$ such that for all $0 < \epsilon < \epsilon_0$ the Laplace–Beltrami operator $-\Delta_{U_\epsilon^{(j)}}$ on the collar satisfies, on the spectral interval $I_\delta := (4j/r^2 + \delta, \infty)$:*

- (i) *A strict Mourre estimate holds with conjugate operator A_j defined initially on $C_c^\infty(U_\epsilon^{(j)})$ by $A_j = \sum_{i=1}^j \frac{1}{2}(y_i \partial_{y_i} + \partial_{y_i} y_i) \chi(y_i \geq Y_0)$ (with cutoff χ supported in the cusp ends), and then extended by closure to an essentially self-adjoint operator on $L^2(U_\epsilon^{(j)})$ (essential self-adjointness follows from the standard theory of first-order symmetric operators generating one-parameter unitary groups: Nelson's analytic vector theorem, Reed–Simon [19, Thm. X.39], with C_c^∞ as the core of analytic vectors for the cusp dilation flow):*

$$E_{I_\delta}(-\Delta_{U_\epsilon^{(j)}})[-\Delta_{U_\epsilon^{(j)}}, iA_j]E_{I_\delta}(-\Delta_{U_\epsilon^{(j)}}) \geq \delta E_{I_\delta}(-\Delta_{U_\epsilon^{(j)}}) + K_\epsilon,$$

with K_ϵ a compact remainder.

- (ii) *A_j is of class $C^{1,1}(-\Delta_{U_\epsilon^{(j)}})$ in the sense of [20, Def. 6.2.2]; equivalently, both the first and second iterated commutators $[-\Delta_{U_\epsilon^{(j)}}, iA_j]$, $[[-\Delta_{U_\epsilon^{(j)}}, iA_j], iA_j]$ extend from C_c^∞ to bounded operators $\mathcal{D}(-\Delta_{U_\epsilon^{(j)}}) \rightarrow \mathcal{D}(-\Delta_{U_\epsilon^{(j)}})^*$ with norm $O(1)$ as $\epsilon \rightarrow 0$.*

- (iii) Consequently [17, Thm. 0.1], $-\Delta_{U_\epsilon^{(j)}}$ has no singular continuous spectrum on I_δ ; its point spectrum on I_δ is locally finite (no accumulation interior to I_δ); and a global limiting absorption principle holds on I_δ , namely the boundary values $\langle A_j \rangle^{-1/2} (-\Delta_{U_\epsilon^{(j)}} - \lambda \mp i0)^{-1} \langle A_j \rangle^{-1/2}$ exist as norm-continuous functions of $\lambda \in I_\delta$.

Proof. For (i): work first on the exact product cusp $\mathcal{M}_{k-j} \times \times_{i=1}^j (\mathbb{H}_r^5\text{-cusp})$. We follow the Froese–Hislop framework [21]: the Mourre identity does *not* hold for the unconjugated upper-half-space Laplacian as an operator identity (direct calculation gives $[-\Delta_{\mathbb{H}_r^5}, iA_i^{(0)}] = 2ir^{-2}y_i^2 \partial_{x^{(i)}}^2$, a transverse-Laplacian-valued operator that vanishes only on the radial sector); the clean Mourre identity instead lives on the *conjugated radial Schrödinger model* obtained by spherical-harmonic decomposition on the horosphere and the standard half-density conjugation.

Concretely, on the i -th cusp factor in upper-half-space coordinates $(x^{(i)}, y_i)$ with $x^{(i)} \in \mathbb{R}^4$, $y_i > 0$, decompose $L^2(\mathbb{H}_r^5\text{-cusp})$ via Fourier transform in the horospherical coordinate $x^{(i)} \in \mathbb{R}^4$ (a flat $\mathbb{R}^4$ at infinity, on which the transverse Laplacian $-\Delta_{x^{(i)}}$ has continuous spectrum $[0, \infty)$). Each Fourier mode at $|k|^2 = \mu \geq 0$ contributes a nonnegative additive shift $\mu \cdot y_i^2/r^2$ to the radial Schrödinger operator below; the $\mu = 0$ (zero-mode) sector is the one we focus on, where the radial reduction is clean. For $\mu > 0$ the additive shift is positive and short-range as $y_i \rightarrow \infty$ (proportional to $y_i^2 \mu$ in the conjugated coordinate $u_i = \log y_i$, becoming $e^{2u_i} \mu \rightarrow \infty$ at $u_i \rightarrow +\infty$), strictly raising the bottom of essential spectrum on those modes, so the threshold $4/r^2$ is set by the zero-mode sector. On the ℓ -th sector the Laplacian, conjugated by the half-density factor $y_i^{(n-1)/2} = y_i^2$ (with $n = 5$) and the substitution $u_i = \log y_i$, becomes a half-line Schrödinger operator on $L^2(\mathbb{R}, du_i)$:

$$H_{\text{conj}}^{(\ell)} = -\partial_{u_i}^2 + \frac{V_\ell(u_i)}{r^2}, \quad V_0(u_i) \equiv 4, \quad V_\ell(u_i) = 4 + \ell(\ell + 3)e^{-2u_i} \quad (\ell \geq 1). \quad (10)$$

The constant $4/r^2$ is the McKean threshold; the Casimir term $\ell(\ell + 3)e^{-2u_i}/r^2$ is short-range as $u_i \rightarrow +\infty$ and provides repulsion on the cusp. On each conjugated sector the dilation generator $A_{u_i} := \frac{1}{2}(u_i D_{u_i} + D_{u_i} u_i)$ with $D_{u_i} = -i\partial_{u_i}$ produces the *exact* Mourre identity on the $\ell = 0$ sector,

$$[H_{\text{conj}}^{(0)}, iA_{u_i}] = 2(-\partial_{u_i}^2) = 2(H_{\text{conj}}^{(0)} - 4/r^2), \quad (11)$$

verified symbolically in [22] (this is the standard half-line dilation–Laplacian commutator). For $\ell \geq 1$ the Casimir term breaks the clean identity but adds a strictly positive contribution: $[H_{\text{conj}}^{(\ell)}, iA_{u_i}] = 2(-\partial_{u_i}^2) - r^{-2}\ell(\ell + 3) \cdot 2u_i e^{-2u_i}$, and on the spectral subspace above the threshold $4/r^2 + \delta$ the short-range Casimir contribution is a relatively compact perturbation of $H_{\text{conj}}^{(\ell)}$ on the cusp end, absorbed into K_ϵ . Transferring back through the unitary conjugation, this gives on the i -th *full* cusp factor the operator identity (modulo a compact remainder from the half-density conjugation and the spherical-harmonic decomposition, both controlled by the compact transverse slice)

$$[-\Delta_{\mathbb{H}_r^5\text{-cusp}, i}, iA_i^{(0)}] = 2(-\Delta_{\mathbb{H}_r^5\text{-cusp}, i} - 4/r^2) + K_{\text{cpt}}^{(i)}, \quad A_i^{(0)} := \frac{1}{2}(y_i \partial_{y_i} + \partial_{y_i} y_i), \quad (12)$$

where $A_i^{(0)}$ is the pullback of A_{u_i} under $u_i = \log y_i$ (the half-density-conjugated form), and $K_{\text{cpt}}^{(i)}$ is the relatively compact transverse-and-Casimir contribution. Summing over i and using the tensor-product decomposition $-\Delta_{\text{prod}} = -\Delta_{\mathcal{M}_{k-j}} \otimes 1 + \sum_i 1 \otimes (-\Delta_{\text{cusp}, i})$:

$$[-\Delta_{\text{prod}}, iA_j^{(0)}] = 2(-\Delta_{\text{prod}} - 4j/r^2) - 2(-\Delta_{\mathcal{M}_{k-j}}) \otimes 1 \geq 2(-\Delta_{\text{prod}} - 4j/r^2) - 2(-\Delta_{\mathcal{M}_{k-j}}). \quad (13)$$

The cutoff $\chi(y_i \geq Y_0)$ introduces a localization error supported on the compact transverse slice $\{y_i = Y_0\}$, which is a relatively compact perturbation of $-\Delta_{\text{prod}}$ (the inclusion $H^1(\text{slice}) \hookrightarrow L^2(\text{slice})$ is compact by Rellich), absorbed into the compact remainder K_ϵ .

Residual factor contribution. The conjugate operator $A_j = \sum_i \frac{1}{2}(y_i \partial_{y_i} + \partial_{y_i} y_i) \chi(y_i \geq Y_0)$ acts *only* on the cusp factors and is the identity on the residual factor $\mathcal{M}_{k-j}(S_r^4)$. In the tensor decomposition $-\Delta_{\text{prod}} = -\Delta_{\mathcal{M}_{k-j}} \otimes 1 + 1 \otimes (-\Delta_{\text{bubbles}})$, the residual Laplacian therefore commutes with A_j :

$$[-\Delta_{\mathcal{M}_{k-j}} \otimes 1, iA_j] = 0$$

identically (not "modulo a compact remainder"). The full commutator $[-\Delta_{\text{prod}}, iA_j] = [1 \otimes (-\Delta_{\text{bubbles}}), iA_j]$ therefore lives entirely on the cusp factors, where the Froese–Hislop identity (12) gives $\sum_i 2(1 \otimes (-\Delta_{\text{cusp},i} - 4/r^2)) + \sum_i K_{\text{cpt}}^{(i)} \otimes 1$. Tensoring with the residual identity, the contribution on the joint spectral subspace $E_{I_\delta}(-\Delta_{\text{prod}})$ is exactly $2(-\Delta_{\text{prod}}|_{\text{cusp}} - 4j/r^2)$ on the cusp factors, which on E_{I_δ} is $\geq 2\delta$ on the cusp-energy part, plus the cumulative compact remainder $K_\epsilon = \sum_i K_{\text{cpt}}^{(i)}$.

The metric perturbation $g_{U_\epsilon^{(j)}} = g_{\text{prod}} + h$ with $\|h\|_{g_{\text{prod}}} \leq C\epsilon |\log \epsilon|$ produces the comparison

$$[-\Delta_{U_\epsilon^{(j)}}, iA_j] = [-\Delta_{\text{prod}}, iA_j^{(0)}] + R_\epsilon^{(1)},$$

with $\pm R_\epsilon^{(1)} \leq C\epsilon |\log \epsilon| (-\Delta_{\text{prod}} + 1)$ as quadratic forms, by Lemma A.7 below applied at order $n = 1$. For $\epsilon < \epsilon_0(\delta)$ small enough this lower-bounds the Mourre constant by δ on I_δ as claimed.

For (ii): the second-iterated commutator

$$[[-\Delta_{U_\epsilon^{(j)}}, iA_j], iA_j] = [[-\Delta_{\text{prod}}, iA_j^{(0)}], iA_j^{(0)}] + R_\epsilon^{(2)}$$

is bounded relative to $-\Delta_{\text{prod}} + 1$ with norm $O(1)$: in the conjugated radial Schrödinger frame (10), the principal symbol of each cusp sector is $-\partial_{u_i}^2$ and the half-line dilation commutator iterates exactly to $[[-\partial_{u_i}^2, iA_{u_i}], iA_{u_i}] = 4(-\partial_{u_i}^2)$, so the unperturbed second commutator equals $4(-\Delta_{\text{prod}} - 4j/r^2)$ modulo a relatively compact remainder (short-range Casimir derivatives and the spherical-harmonic compactness, as in (i)); $R_\epsilon^{(2)} = O(\epsilon |\log \epsilon|^2)$ by Lemma A.7 at $n = 2$. This is the $C^{1,1}(-\Delta_{U_\epsilon^{(j)}})$ regularity needed by Sahbani [17, Hyp. H1–H3, p. 411]; in fact one obtains the strictly stronger $C^2(-\Delta_{U_\epsilon^{(j)}})$ regularity.

Statement (iii) is then the conclusion of [17, Thm. 0.1] (or equivalently [20, Thm. 7.5.4] together with the standard absence of σ_{sc} corollary). $\square$

Remark 4.9 (On the absolute-continuity strengthening). The conjunction of statements (i)–(iii) of Theorem 4.8 implies that the spectrum of $-\Delta_{U_\epsilon^{(j)}}$ on I_δ is the union of (a) absolutely continuous spectrum filling I_δ and (b) at most finitely many embedded eigenvalues on I_δ for each compact sub-interval (locally finite point spectrum). Whether embedded eigenvalues are present at all on I_δ is not settled by Mourre theory and depends on the residual moduli factor; on the model cusp $\mathbb{H}_r^5$ itself the spectrum is purely absolutely continuous and no eigenvalues appear.

Remark 4.10 ($\text{SU}(N)$ extension). Theorem 4.3 holds verbatim for the moduli space $\mathcal{M}_k(S_r^4; \text{SU}(N))$ of charge- k anti-self-dual $\text{SU}(N)$ connections, $N \geq 2$, with the same collar essential bottom $4j/r^2$. The η -tensor identity used in Lemma 4.1 operates entirely inside the $\mathfrak{su}(2)$ -subalgebra in which each bubble is framed, and the canonical normalization $\text{Tr}(T_a T_b) = \frac{1}{2} \delta_{ab}$ makes the $\text{SU}(N)$ common-subgroup cross-term identical to (5). Bubbles framed in distinct $\text{SU}(2)$ subgroups of $\text{SU}(N)$, related by $U \in \text{SU}(N)$, acquire a framing-overlap scalar

$$o(U) := \frac{2}{3} \sum_{a=1}^3 \text{Tr}_{\text{fund}}(T_a U T_a U^{-1}) \in [-1/3, 1],$$

where $\{T_1, T_2, T_3\}$ is the fixed basis of the reference $\mathfrak{su}(2) \subset \mathfrak{su}(N)$ subalgebra in which the first bubble is framed (with the canonical normalization $\text{Tr}_{\text{fund}}(T_a T_b) = \frac{1}{2} \delta_{ab}$), and U is the relative framing rotation to the second bubble's $\mathfrak{su}(2)$ subalgebra. The factor $2/3$ normalizes o so that the same-subgroup case $U = 1$ gives $o(1) = (2/3) \cdot 3 \cdot \frac{1}{2} = 1$, recovering the $\text{SU}(2)$ value; the value $-1/3$ is attained on the symmetric self-conjugate fundamental representation. In all cases the factor $o(U)$ preserves the $O(s_1 s_2 / R^2)$ decay rate. The additional compact framing factors $\text{SU}(N)/S(U(N-2) \times U(2))$ on each bubble are positively curved and contribute only discrete spectrum starting at 0 to each cusp; they do not lower the essential bottom. Symbolic verification of the $\text{SU}(N)$ cross-term and numerical confirmation at $N = 2, 3, 4$ are recorded in [10].

5 Remarks and open questions

Remark 5.1 (Sign of Ric on instanton moduli). $\mathbb{H}_r^5$ has $\text{Ric} = -4g/r^2$, strictly negative. The Bakry–Émery–Lichnerowicz framework with positive Ricci would give a discrete eigenvalue gap; with negative Ricci, it gives a continuous spectrum with positive bottom, consistent with Theorem 3.2 and Theorem 4.3. Singer's orbit-space proposal [1] requires positive Ricci on the relevant sector for Lichnerowicz–Obata to give a spectral gap; the instanton sectors carry continuous spectrum bounded by McKean rather than discrete eigenvalues.

Remark 5.2 (Open: global $\lambda_0(\mathcal{M}_k)$ for $k \geq 2$). Theorem 4.3 bounds the essential-spectrum bottom from each collar but does not determine the global $\lambda_0(\mathcal{M}_k)$ for $k \geq 2$. The natural attack uses the explicit ADHM moduli geometry and the Groisser–Parker interior curvature bounds, combined with a Bargmann-type argument on the radial reduction in the $\text{SO}(5)$ -invariant cohomogeneity-one sector.

Remark 5.3 (Additional structural results in the companion supplementary note). Two structural reductions of the global $\lambda_0(\mathcal{M}_2)$ question, an $\text{SO}(5)$ -isotypic decomposition reducing the question to the invariant subspace via Bargmann + Casimir repulsion on the $\ell \geq 1$ components, and a Mazzeo–Melrose-type finiteness-and-meromorphic-resolvent statement for $\sigma_{\text{pp}} \cap (0, 4/r^2)$ via iterated-edge 0-calculus on stratified asymptotically hyperbolic manifolds, are recorded as conditional propositions in the companion supplementary note. Each rests on inputs (uniform-in-family Bargmann robustness for the $\text{SO}(5)$ -isotypic reduction; stratified iterated-edge calculus verification for the 0-calculus statements) whose self-contained derivations are beyond the scope of this note.

A Weighted-Sobolev proof of Lemma 4.2(iv)–(v)

This appendix proves the two off-diagonal cross-block bounds of Lemma 4.2 that were stated in the body without a self-contained calculation. The argument is the standard Taubes [7]/Donaldson–Kronheimer [12] §7.3 weighted-Sobolev framework, with the scale-uniformity made explicit. We cite Bartnik [8] and Lockhart–McOwen [11] for the abstract Fredholm theory of the weighted Laplacian on $\mathbb{R}^n$; everything else is proved here.

A.1 Weighted Sobolev setup

On $\mathbb{R}^4$, fix a smooth weight $\sigma(x) = (1 + |x - x_0|^2)^{1/2}$ and for $\beta \in \mathbb{R}$, define

$$\|f\|_{L_\beta^2}^2 := \int_{\mathbb{R}^4} |f(x)|^2 \sigma(x)^{-2\beta-4} d^4x, \quad \|f\|_{W_\beta^{2,2}}^2 := \sum_{|\alpha| \leq 2} \|\sigma^{|\alpha|} \nabla^\alpha f\|_{L_\beta^2}^2,$$

following the Bartnik convention [8]: a function with pointwise rate $|f(x)| \leq C \sigma(x)^{\beta'}$ lies in L_β^2 for $\beta' < \beta$, and the borderline rate $|f| \sim \sigma^{\beta'}$ is in L_β^2 iff $\beta' < \beta$ (strict). The pairing $L_\beta^2 \times L_{-4-\beta}^2 \rightarrow \mathbb{R}$ identifies $(L_\beta^2)^* \cong L_{-4-\beta}^2$ [8]. *Convention note:* this is the Bartnik sign convention; the Lockhart–McOwen weight convention [11] uses the opposite sign for β . Both yield the same Fredholm theory after a sign flip; we work throughout with the Bartnik convention, in which $\Delta : W_\beta^{2,2} \rightarrow L_{\beta-2}^2$ on $\mathbb{R}^4$ is Fredholm for non-critical β (the critical exponents are the integers $\beta \in \{-2, -3, -4, \dots\}$ corresponding to the kernel/cokernel obstructions of the Laplacian on the asymptotically Euclidean half-line).

We work with the weight $\beta = -3/2$. This is the natural weight for the BPST tangent space because:

- The field-strength rate $|F_A(x)| = O(s^2/|x - x_0|^4)$ lies in L_β^2 for any $\beta > -4$.
- The bare position-derivative tail $|v_{x,\rho}(x)| = O(|x - x_0|^{-2})$ fails to lie in plain $L^2 = L_{-2}^2$ (it sits exactly at the critical exponent $\beta_{\text{crit}} = -2$); however, by the Bartnik strict-inequality criterion above, $v_{x,\rho} \in L_\beta^2$ for every $\beta > -2$, in particular for the working weight $\beta = -3/2$, with norm growing like $|\log s|^{1/2}$ as $s \rightarrow 0$. The Coulomb obstruction is therefore that $v_{x,\rho}$ does not lie in the *energy space* (in which derivatives are also weighted) without renormalization. The horizontal projection $\hat{v}_{x,\rho}$ instead satisfies $|\hat{v}_{x,\rho}| = O(s^2/(r+s)^3)$ (Lemma A.2) and belongs to $L_{-3/2}^2$ with norm uniformly bounded in s .
- The horizontal-projected rate $|\hat{v}_{x,\rho}(x)| = O(s^2/|x - x_0|^3)$ is strictly inside $L_{-3/2}^2$, with L_β^2 -norm uniformly bounded in s .

On the BPST background A with scale s , write $\Delta_A = d_A^* d_A$ acting on $\mathfrak{g}$ -valued functions. The conformal/scale invariance of the BPST connection under $x \mapsto x_0 + s x'$, $A \mapsto s^{-1} A'$ rescales $\Delta_A \mapsto s^{-2} \Delta_{A'}$, and rescales the weighted norms by $\|f\|_{L_\beta^2} \mapsto s^{2+\beta} \|f'\|_{L_\beta^2}$ (the factor s^2 from the volume Jacobian, plus s^β from the weight). The combination is what produces the scale-uniform constants below.

Lemma A.1 (Scale-uniform weighted Coulomb estimate). *There exists C_0 , independent of the BPST scale s , such that for every $\mathfrak{g}$ -valued 1-form v on $\mathbb{R}^4$ with $d_A^* v \in L_{-1/2}^2$, the equation*

$$\Delta_A \phi = d_A^* v$$

admits a unique solution $\phi \in L_{-3/2}^2$ satisfying

$$\|\phi\|_{W_{-3/2}^{2,2}} \leq C_0 \|d_A^* v\|_{L_{-1/2}^2}.$$

Proof. For $A \equiv 0$ (the flat background), the weight $\beta = -3/2$ is non-critical for Δ on $\mathbb{R}^4$ in the sense of Lockhart–McOwen [11], lying strictly between the critical exponents $\beta = -2$ (kernel: constants) and $\beta = -4$ (cokernel obstruction). Bartnik’s theorem [8, Thm. 1.7] then states that $\Delta : W_{-3/2}^{2,2} \rightarrow L_{-1/2}^2$ is an isomorphism, giving an estimate $\|\phi\|_{W_{-3/2}^{2,2}} \leq C_1 \|\Delta \phi\|_{L_{-1/2}^2}$ with C_1 an absolute constant. *Convention reconciliation:* the Bartnik convention adopted here pairs a domain weight $\beta = -3/2$ with target weight $-1/2$ via the duality $(L_{-1/2}^2)^* \cong L_{-7/2}^2$; the equivalent alternative convention $\Delta : W_\beta^{2,2} \rightarrow L_{\beta-2}^2$ yields target weight $-7/2$ at $\beta = -3/2$. Both conventions give identical Fredholm content; we work throughout with the Bartnik form.

For the BPST background at scale $s = 1$ centered at $x_0 = 0$, write $\Delta_A = \Delta + [A, \cdot]$ where the second term is a first-order operator with coefficient $|A(x)| = O(|x|/(1+|x|^2)) = O(\sigma^{-1})$ pointwise.

The perturbation $K := \Delta_A - \Delta = 2A \cdot \nabla + (\nabla \cdot A) + A \cdot A$ has $K : W_{-3/2}^{2,2} \rightarrow L_{-1/2}^2$ compact (the coefficients decay like σ^{-1} , σ^{-2} , σ^{-2} respectively, gaining one full unit of decay relative to the borderline). Triviality of the kernel of Δ_A on $\mathfrak{g}$ -valued functions in $L_{-3/2}^2$ is the AHS/Taubes infinitesimal-deformation vanishing: any $\phi \in L_{-3/2}^2$ with $\Delta_A \phi = 0$ satisfies $\langle \phi, \Delta_A \phi \rangle_{L^2} = \|d_A \phi\|_{L^2}^2 = 0$ (the integration by parts is justified by the strict $\sigma^{-3/2}$ decay), so $d_A \phi = 0$; on an irreducible bundle the only covariantly constant $\mathfrak{g}$ -valued function is $\phi \equiv 0$ [9, Prop. 3.4] (see also [12, Lem. 4.2.4]). Hence $\Delta_A : W_{-3/2}^{2,2} \rightarrow L_{-1/2}^2$ is invertible, with an estimate $\|\phi\|_{W_{-3/2}^{2,2}} \leq C_0 \|\Delta_A \phi\|_{L_{-1/2}^2}$.

We now extract the s -uniformity by conformal scaling. Under the dilation $\Phi_s : x' \mapsto x_0 + sx'$, the BPST connection at scale s is the pullback (up to a constant gauge transformation) of the unit-scale BPST connection: $\Phi_s^* A_s = A_{s=1}$, so $\Delta_{A_s} = s^{-2} \Phi_s^* \Delta_{A_{s=1}}$. Track the scalings on each side of $\|\phi\|_{W_{\beta}^{2,2}(\mathbb{R}^4)} \leq C \|d_A^* v\|_{L_{\beta+1}^2(\mathbb{R}^4)}$:

- Volume element: $d^4 x = s^4 d^4 x'$.
- Weight $\sigma(x) = (1 + |x|^2)^{1/2}$ at large $|x|$: $\sigma(x) \sim s \sigma(x')$, so $\sigma^{-2\beta-4} d^4 x \sim s^{-2\beta-4} \cdot s^4 d^4 x' = s^{-2\beta} d^4 x'$.
- L_{β}^2 norm of ϕ scales by $s^{-\beta+0}$ (after taking square root): $\|\phi\|_{L_{\beta}^2(\mathbb{R}_x^4)} = s^{-\beta} \|\phi'\|_{L_{\beta}^2(\mathbb{R}_{x'}^4)}$ where $\phi'(x') = \phi(\Phi_s(x'))$.
- Two derivatives (∂^2 adds factor s^{-2} in the operator, while integration by parts in the norm-squared form gives a factor s^{+2} extra in the norm): $\|\phi\|_{W_{\beta}^{2,2}} \asymp s^{-\beta-2+\text{volume}} \cdot \|\phi'\|_{W_{\beta}^{2,2}}$. The full bookkeeping gives $\|\phi\|_{W_{\beta}^{2,2}(\mathbb{R}_x^4)} = s^{-\beta+2} \|\phi'\|_{W_{\beta}^{2,2}(\mathbb{R}_{x'}^4)}$ at $\beta = -3/2$.
- Similarly $\|d_A^* v\|_{L_{\beta+1}^2}$ scales by $s^{-(\beta+1)+2} = s^{-\beta+1}$ (one derivative loses s^{-1} relative to the norm; the additional -1 in the weight shift to $\beta+1$ adds another).

The two sides scale by the same total power $s^{-\beta+2} = s^{7/2}$ at $\beta = -3/2$, so the inequality $\|\phi\|_{W_{\beta}^{2,2}} \leq C \|d_A^* v\|_{L_{\beta+1}^2}$ is invariant under Φ_s , and the constant C_0 obtained at $s = 1$ transfers verbatim to all $s > 0$. $\square$

A.2 Tail decay of the horizontal projection

Let A be a BPST instanton centered at x_0 with scale s , and let $v_{x,\rho} = \partial A / \partial x_0^\rho$ be the bare position-derivative tangent vector. We work in the regular gauge $A_\mu^a = 2\eta_{\mu\nu}^a (x - x_0)^\nu / D$, $D := |x - x_0|^2 + s^2$.

Lemma A.2 (Coulomb source and pointwise tail). *The horizontal correction ϕ^ρ defined by $\Delta_A \phi^\rho = d_A^* v_{x,\rho}$ satisfies, with $r := |x - x_0|$,*

$$|\phi^\rho(x)| \leq C \frac{s^2}{r+s}, \quad |d_A \phi^\rho(x)| \leq C \frac{s^2}{(r+s)^2},$$

and hence $\hat{v}_{x,\rho} := v_{x,\rho} - d_A \phi^\rho$ satisfies the field-strength rate

$$|\hat{v}_{x,\rho}(x)| \leq C \frac{s^2}{(r+s)^3}.$$

In particular $\|\hat{v}_{x,\rho}\|_{L^2(\mathbb{R}^4)} \leq C$ uniformly in s , and $\|\hat{v}_{x,\rho}\|_{L_{\beta}^2} \leq C$ for any $\beta > -3/2$ (Bartnik weight).

Proof. A direct calculation in the regular gauge gives $d_A^* v_{x,\rho} = -2\eta_{\mu\rho}^a \partial^\mu (1/D)$ (the BPST identity $\Delta A = 0$ in this gauge implies the only surviving term comes from differentiating the explicit x_0 -dependence in D). The pointwise rate is $|d_A^* v_{x,\rho}(x)| = O(s^2(r+s)/D^2) = O(s^2/(r+s)^3)$. Computing the weighted norm,

$$\|d_A^* v_{x,\rho}\|_{L_{-1/2}^2}^2 = \int \frac{s^4}{(r+s)^6} (1+r^2)^{-3/2-2} d^4x = O(s^4) \int_0^\infty \frac{r^3 dr}{(r+s)^6(1+r)^7} \leq C s^4 \cdot s^{-4} = C$$

(both inner ($r \sim s$) and outer regions contribute $O(1)$ after the s^4 prefactor; explicit scaling $r = s\rho$ inner, r outer). Thus $d_A^* v_{x,\rho} \in L_{-1/2}^2$ uniformly in s , with norm $\|d_A^* v_{x,\rho}\|_{L_{-1/2}^2} = O(1)$.

Lemma A.1 gives $\phi^\rho \in W_{-3/2}^{2,2}$ with $\|\phi^\rho\|_{W_{-3/2}^{2,2}} \leq C_0 \cdot O(1) = O(1)$, uniformly in s . The pointwise bound on ϕ^ρ now follows from the standard interior-elliptic-regularity bootstrap combined with the explicit form of the source: $d_A^* v_{x,\rho}$ is $O(s^2(r+s)^{-3})$, and convolution with the $\mathbb{R}^4$ Newton kernel $\sim |y|^{-2}$ produces $|\phi^\rho| = O(s^2/(r+s))$. Differentiating once gives $|d_A \phi^\rho| = O(s^2/(r+s)^2)$.

We now identify the cancellation explicitly. Translations $x_0 \mapsto x_0 + \epsilon e_\rho$ are gauge equivalences of the BPST family: the connection $A(x; x_0 + \epsilon e_\rho)$ is gauge-equivalent to $A(x; x_0)$ via a smooth gauge transformation $g_\epsilon \rightarrow 1$ as $\epsilon \rightarrow 0$, so the infinitesimal translation $v_{x,\rho} = \partial A / \partial x_0^\rho$ equals an infinitesimal gauge transformation $-d_A \psi^\rho$ up to a non-pure-gauge piece. Writing the gauge transformation as $\psi^\rho \in \Omega^0(\mathbb{R}^4; \mathfrak{g})$, the Bianchi identity $d_A F_A = 0$ implies that the non-pure-gauge piece $\hat{v}_{x,\rho} = v_{x,\rho} + d_A \psi^\rho$ contracts only with the field-strength tensor, i.e. inherits the curvature decay rate $|F_A| = O(s^2(r+s)^{-4})$ multiplied by a length factor. The Coulomb-projected ϕ^ρ is precisely the unique gauge transformation ψ^ρ that makes $\hat{v}_{x,\rho}$ Coulomb-horizontal ($d_A^* \hat{v}_{x,\rho} = 0$); existence and uniqueness are Lemma A.1. Hence

$$|\hat{v}_{x,\rho}(x)| = |v_{x,\rho} - d_A \phi^\rho| = O(s^2(r+s)^{-3})$$

pointwise, the leading $|x - x_0|^{-2}$ gauge tails of $v_{x,\rho}$ and of $d_A \phi^\rho$ canceling exactly. The L^2 norm is $\|\hat{v}_{x,\rho}\|_{L^2}^2 = O(s^4) \int r^3 dr / (r+s)^6 = O(1)$. $\square$

A.3 Green's-kernel pointwise estimate for the BPST horizontal projection

The pointwise bound $|\hat{v}_{x,\rho}(x)| = O(s^2/(r+s)^3)$ of Lemma A.2 was deduced from the Bianchi-identity / gauge-equivalence argument, which is structural. Following the requests of the external review, we record here the same bound via the explicit Green's-kernel representation of the BPST-background Laplacian, which makes the cancellation of the long-range $|x - x_0|^{-2}$ tail manifest. The argument is standard (Taubes [7] §4; Donaldson–Kronheimer [12] §7.3) but we spell it out for transparency.

Lemma A.3 (Green's-kernel pointwise estimate for the BPST horizontal projection). *Let A be a BPST instanton on $\mathbb{R}^4$ centered at x_0 with scale s , and let $G_A(x, x')$ denote the integral kernel of the Green's operator Δ_A^{-1} for the BPST-background Laplacian on $\mathfrak{g}$ -valued 0-forms, in the Bartnik weight class $\beta = -3/2$ (Lemma A.1). Write $G_0(x, x') = (4\pi^2)^{-1} |x - x'|^{-2}$ for the bare $\mathbb{R}^4$ Newton kernel. Then:*

- (a) $G_A(x, x') = G_0(x, x') + R_A(x, x')$ with the remainder satisfying the pointwise bound

$$|R_A(x, x')| \leq \frac{C}{(1 + |x - x'|)^2 (1 + |x - x_0|) (1 + |x' - x_0|)},$$

uniformly in the scale s in the Bartnik–Lockhart–McOwen weight class.

(b) The horizontal correction $\phi^\rho = G_A d_A^* v_{x,\rho}$ satisfies the explicit kernel representation

$$\phi^\rho(x) = \int_{\mathbb{R}^4} G_A(x, x') d_A^* v_{x,\rho}(x') d^4 x',$$

and the pointwise bound $|d_A \phi^\rho(x)| \leq C s^2 / (r + s)^2$ of Lemma A.2 is obtained by direct integration against the explicit source $|d_A^* v_{x,\rho}(x')| = O(s^2 / (|x' - x_0| + s)^3)$.

(c) The leading $|x - x_0|^{-2}$ tail of the bare $v_{x,\rho}$ and the leading $|x - x_0|^{-2}$ tail of $d_A \phi^\rho$ cancel pointwise, leaving the field-strength rate

$$|\hat{v}_{x,\rho}(x)| \leq \frac{C s^2}{(|x - x_0| + s)^3}$$

uniformly in s and x . The constant C depends only on the BPST profile.

Proof. (a) On $\mathbb{R}^4$ the bare Laplacian Green's function is $G_0(x, x') = (4\pi^2)^{-1} |x - x'|^{-2}$. The BPST background contributes a perturbation

$$\Delta_A - \Delta_0 = -[A^\mu, \partial_\mu \cdot] - \partial^\mu [A_\mu, \cdot] - [A^\mu, [A_\mu, \cdot]],$$

with coefficients of pointwise size $|A| = O(s / (r + s)^2)$ and $|A|^2 = O(s^2 / (r + s)^4)$. In the Bartnik weight $\beta = -3/2$ class, Lemma A.1 establishes that $\Delta_A : W_{-3/2}^{2,2} \rightarrow L_{-1/2}^2$ is an isomorphism uniformly in s , so its inverse Δ_A^{-1} has an integral kernel G_A admitting the resolvent series

$$G_A = G_0 - G_0 (\Delta_A - \Delta_0) G_0 + G_0 (\Delta_A - \Delta_0) G_0 (\Delta_A - \Delta_0) G_0 - \cdots,$$

convergent in the weighted operator-norm topology by the scale-uniform Coulomb estimate. The first-order remainder is bounded by

$$|R_A^{(1)}(x, x')| \leq \int |G_0(x, y)| |A(y)| |\partial G_0(y, x')| d^4 y + (\text{lower order}),$$

and the weighted Riesz convolution estimate $\int |x - y|^{-2} (1 + |y|)^{-2} (1 + |y|)^{-2} |y - x'|^{-3} d^4 y \leq C(1 + |x - x'|)^{-2} (1 + |x|)^{-1}$ on $\mathbb{R}^4$, a Hardy–Littlewood–Sobolev / Riesz-composition statement (sum of homogeneity orders $2 + 3 = 5 > 4 = \dim$ giving the extra factor $(1 + |x|)^{-1}$); see Stein [18], Ch. V, Thm. 1, gives the pointwise bound. Higher-order terms gain further factors of $(1 + |x - x_0|)^{-1}$, so they are subleading. The translation-invariance argument may be reformulated as: $G_A - G_0 = G_0 V G_A$ with $V = \Delta_A - \Delta_0$, and a direct Schur test in the Bartnik weight closes the iteration.

(b) Direct substitution: $\Delta_A \phi^\rho = d_A^* v_{x,\rho}$ inverts to $\phi^\rho(x) = \int G_A(x, x') d_A^* v_{x,\rho}(x') d^4 x'$. Split $G_A = G_0 + R_A$:

$$\phi^\rho(x) = \underbrace{\int G_0(x, x') d_A^* v_{x,\rho}(x') d^4 x'}_{=: \phi_0^\rho(x)} + \int R_A(x, x') d_A^* v_{x,\rho}(x') d^4 x'.$$

The first integral is the classical $\mathbb{R}^4$ Newton-kernel convolution of the source $d_A^* v_{x,\rho}$, which by the regular-gauge identity $d_A^* v_{x,\rho} = -2\eta_{\mu\rho}^a \partial^\mu (1/D)$ (proof of Lemma A.2) telescopes to

$$\phi_0^\rho(x) = -2\eta_{\mu\rho}^a \partial^\mu \int G_0(x, x') D(x')^{-1} d^4 x' = -2\eta_{\mu\rho}^a \partial^\mu \left[\frac{1}{4\pi^2} \int |x - x'|^{-2} (|x' - x_0|^2 + s^2)^{-1} d^4 x' \right],$$

and the inner integral is the standard four-dimensional convolution of two Yukawa-type kernels, evaluated in closed form: $(4\pi^2)^{-1} \int |x - x'|^{-2} (|x' - x_0|^2 + s^2)^{-1} d^4 x' = (4\pi^2)^{-1} \pi^2 r^{-1} \arctan(r/s)$ with $r = |x - x_0|$. Differentiating once: $\partial^\mu \phi_0^\rho \sim -2\eta_{\mu\rho}^a (x - x_0)^\mu / D + O(s^2/(r+s)^3)$, so that $d_A \phi_0^\rho$ has a leading $|x - x_0|^{-2}$ tail that matches the leading tail of $v_{x,\rho}$ pointwise. The second integral, involving R_A , contributes only $O(s^2/(r+s)^3)$ by part (a) and the source decay.

(c) Subtracting,

$$\hat{v}_{x,\rho}(x) = v_{x,\rho}(x) - d_A \phi_0^\rho(x) = [v_{x,\rho}(x) - d_A \phi_0^\rho(x)] - d_A \int R_A(x, x') d_A^* v_{x,\rho}(x') d^4 x'.$$

The bracketed difference is the bare Coulomb projection in the *flat* background, which by the analogous Bianchi identity for the abelian-leading part (or equivalently, by the closed-form $\arctan(r/s)$ evaluation above) has its leading $|x - x_0|^{-2}$ tail cancelled exactly, leaving $|v_{x,\rho} - d_A \phi_0^\rho| = O(s^2/(r+s)^3)$. The R_A -tail integral contributes a further $O(s^2/(r+s)^3)$ by (a) and (b). Combining the two estimates gives the claimed pointwise bound $|\hat{v}_{x,\rho}(x)| \leq C s^2/(r+s)^3$, uniformly in s . This recovers Lemma A.2 via the explicit kernel route, as required by the external review. $\square$

A.4 Two-bubble cross-block bound

Proposition A.4 (Proof of Lemma 4.2(iv)). *Let A_1, A_2 be two BPST instantons centered at x_1, x_2 with $R = |x_1 - x_2|$, scales $s_1, s_2 \leq cR$ for some $c < 1$, and let $\hat{v}_\alpha^{(i)}$ be the gauge-fixed tangent vectors ($\alpha \in \{s, x_1, \dots, x_4\}$). Then*

$$|\langle \hat{v}_\alpha^{(1)}, \hat{v}_\beta^{(2)} \rangle_{L^2(\mathbb{R}^4)}| \leq \frac{C s_1 s_2}{R^2} \|\hat{v}_\alpha^{(1)}\|_{L^2} \|\hat{v}_\beta^{(2)}\|_{L^2} (1 + |\log(\min(s_1, s_2)/R)|),$$

with C depending only on c .

Proof. By Lemma A.2, $|\hat{v}_\alpha^{(i)}(x)| \leq C s_i^2/(|x - x_i| + s_i)^3$ for $\alpha \in \{x_1, \dots, x_4\}$ (the position directions). For $\alpha = s$, $v_s^{(i)} = \partial A / \partial s$ already has the field-strength rate $|v_s^{(i)}| = O(s_i/(|x - x_i| + s_i)^2) \cdot s_i = O(s_i^2/(|x - x_i| + s_i)^3)$ without any gauge correction needed (it is naturally Coulomb-horizontal up to a Coulomb shift of the same scale). In either case

$$|\hat{v}_\alpha^{(i)}(x) \hat{v}_\beta^{(j)}(x)| \leq \frac{C s_1^2 s_2^2}{(|x - x_1| + s_1)^3 (|x - x_2| + s_2)^3}.$$

Split the integral over three regions: $\Omega_1 = \{|x - x_1| \leq R/3\}$, $\Omega_2 = \{|x - x_2| \leq R/3\}$, $\Omega_3 = \mathbb{R}^4 \setminus (\Omega_1 \cup \Omega_2)$.

Region Ω_1 . On $\Omega_1 = \{|x - x_1| \leq R/3\}$ we have $|x - x_2| \geq 2R/3$, so pointwise

$$|\hat{v}_\alpha^{(1)}(x)| \leq \frac{C s_1^2}{(|x - x_1| + s_1)^3}, \quad |\hat{v}_\beta^{(2)}(x)| \leq \frac{C s_2^2}{R^3}.$$

The second factor is constant on Ω_1 and pulls out. Writing $r = |x - x_1|$ and substituting $r = s_1 \rho$,

$$\int_{\Omega_1} \frac{s_1^2 d^4 x}{(r + s_1)^3} = 2\pi^2 s_1^2 \int_0^{R/3} \frac{r^3 dr}{(r + s_1)^3} = 2\pi^2 s_1^3 \int_0^{R/(3s_1)} \frac{\rho^3 d\rho}{(1 + \rho)^3} \leq C s_1 R^2,$$

using $\int_0^M \rho^3 d\rho / (1 + \rho)^3 \leq \frac{1}{2} M^2$ for $M \geq 1$. Multiplying by the Ω_1 value $C s_2^2 / R^3$ of the second factor,

$$\int_{\Omega_1} |\hat{v}_\alpha^{(1)} \hat{v}_\beta^{(2)}| d^4 x \leq C s_1 R^2 \cdot \frac{s_2^2}{R^3} = C \frac{s_1 s_2^2}{R}.$$

Using the hypothesis $s_2 \leq cR$ twice, $s_1 s_2^2 / R = (s_1 s_2 / R^2) \cdot s_2 R \leq c \cdot (s_1 s_2 / R^2) \cdot R^2 \leq c R^2 \cdot (s_1 s_2 / R^2)$. Since R is bounded above by the diameter of the collar region (which is itself bounded by the diameter of S_r^4), the factor R^2 is absorbed into an absolute constant $C(c, r)$, giving $\int_{\Omega_1} |\hat{v}_\alpha^{(1)} \hat{v}_\beta^{(2)}| d^4 x \leq C(c, r) \cdot s_1 s_2 / R^2$. Combined with $\|\hat{v}_\alpha^{(i)}\|_{L^2} = \Theta(1)$ (Lemma A.2), this is the target bound for the Ω_1 contribution.

Region Ω_2 . The symmetric calculation gives $\int_{\Omega_2} |\hat{v}^{(1)} \hat{v}^{(2)}| d^4 x \leq C s_1^2 s_2 / R$, absorbed by the same argument with the roles of s_1 and s_2 swapped.

Region Ω_3 (far/annular). Here $|x - x_i| \geq R/3$ for both i ; the pointwise bound $|\hat{v}^{(i)}| \leq C s_i^2 / |x - x_i|^3$ applies, so

$$\int_{\Omega_3} \frac{s_1^2 s_2^2 d^4 x}{|x - x_1|^3 |x - x_2|^3} \leq C s_1^2 s_2^2 \int_{R/3}^\infty \frac{r^3 dr}{r^6} = C \frac{s_1^2 s_2^2}{R^2}.$$

Since $s_1 s_2 / R^2 \leq c^2$, this is $\leq C s_1 s_2 / R^2 \cdot s_1 s_2 / R^2 \cdot R^2 \leq C s_1 s_2 / R^2$ after absorbing the $\|\hat{v}^{(i)}\|_{L^2} = \Theta(1)$ normalisation.

Case split for the log. For index pairs (α, β) such that at least one of the tangent vectors is isotropic (i.e. $\alpha = s$ or $\beta = s$, or the position direction is parallel to the bubble axis $x_1 x_2$), the modulus bounds above give the stated estimate without any logarithm: combining $\Omega_1, \Omega_2, \Omega_3$ yields $|\langle \hat{v}_\alpha^{(1)}, \hat{v}_\beta^{(2)} \rangle| \leq C s_1 s_2 / R^2$.

The logarithm arises only when *both* α, β are transverse position directions, $\alpha = x_\rho, \beta = x_\sigma$ with $\rho, \sigma \in \{2, 3, 4\}$ (orthogonal to the bubble separation axis). In this dipole-dipole case the modulus bounds above are not sharp. The matched-asymptotic expansion of $\hat{v}_{x,\rho}^{(i)}$ in the intermediate annulus $s_i \ll |x - x_i| \ll R$ comes from Lemma A.2 applied to the leading Coulomb-projected term: in regular gauge the bare $v_{x,\rho}^{(i)}$ has the quadrupole-type form $-2\eta_{\mu\rho}^a / D_i + 4\eta_{\mu\nu}^a \xi^{(i),\nu} \xi^{(i),\rho} / D_i^2$ where $\xi^{(i)} := x - x_i$, and after subtracting $d_A \phi^{\rho,(i)}$ which removes the $|x - x_i|^{-2}$ tail (cf. Lemma A.2), the residual in the intermediate annulus reads

$$\hat{v}_{x,\rho}^{(i),\mu,a}(x) = s_i^2 T_\rho^{\mu,a}(\hat{\xi}^{(i)}) / |x - x_i|^4 + O(s_i^4 / |x - x_i|^6), \quad \hat{\xi}^{(i)} := \xi^{(i)} / |\xi^{(i)}|,$$

where the explicit traceless tensor $T_\rho^{\mu,a}(\hat{\xi}) = 4\eta_{\mu\nu}^a \hat{\xi}^\nu \hat{\xi}^\rho - \eta_{\mu\rho}^a$ (which satisfies $\eta_{\mu\nu}^a \delta^{\nu\rho} \hat{\xi}_\rho = 0$ on average and has traceless angular content by the η -tensor symmetries). The dipole-dipole pairing on Ω_3 reads

$$s_1^2 s_2^2 \int_{\Omega_3} \frac{T_\rho(x - x_1) \cdot T_\sigma(x - x_2)}{|x - x_1|^4 |x - x_2|^4} d^4 x,$$

the dot denoting contraction in the spin-color index (μ, a) . Setting $x = Ry$ rescales this to $s_1^2 s_2^2 R^{-4} \int K(y) dy$ on the dilated domain $\{|y - y_i| \geq 1/3, |y - y_i| \geq s_i / R\}$. The kernel $K(y) = T_\rho(y - \hat{y}_1) \cdot T_\sigma(y - \hat{y}_2) / |y - y_1|^4 |y - y_2|^4$ is integrable at infinity ($\int_1^\infty r^3 dr / r^8 = O(1)$); near each pole $y \rightarrow y_i$, the angular integral of $T_\rho(y - \hat{y}_i)$ against any vector field constant in y vanishes (tracelessness of T_ρ in $\hat{\xi}$), so the would-be $|y - y_i|^{-4} \cdot r^3$ radial singularity is improved by one power to $|y - y_i|^{-3} \cdot r^3 = O(1/r)$ as $r = |y - y_i| \rightarrow 0$. The radial part contributes

$$\int_{s_i/R}^{1/3} \frac{dr}{r} = \log(R / (3s_i)) = O(\log(R / \min(s_1, s_2))).$$

Hence the dipole-dipole pairing is bounded by $C s_1^2 s_2^2 R^{-4} \log(R / \min(s_i)) = C (s_1 s_2 / R^2) \cdot (s_1 s_2 / R^2) \log(R / \min(s_i))$. Using $s_1 s_2 / R^2 \leq c^2$ absorbs one factor of $s_1 s_2 / R^2$ into a constant, yielding the stated bound with the $|\log(\min(s_i) / R)|$ enhancement.

Combining the case split: for all index pairs (α, β) the bound $C (s_1 s_2 / R^2) (1 + |\log(\min(s_i) / R)|) \|\hat{v}_\alpha^{(1)}\|_{L^2} \|\hat{v}_\beta^{(2)}\|_{L^2}$ holds. $\square$

A.5 Bubble-background bound (Lemma 4.2(v))

Proposition A.5 (Proof of Lemma 4.2(v)). *Let a be a tangent vector to $\mathcal{M}_{k-j}$. In the standard Taubes–Donaldson–Kronheimer Uhlenbeck-gluing recipe ([7]; [12, §7.2.1]), a is realized on S_r^4 by a cutoff: a smooth function $\beta_i \in C_c^\infty(S_r^4 \setminus B(x_i, R/2))$ with $\beta_i \equiv 1$ outside $B(x_i, R)$ multiplies the $\mathcal{M}_{k-j}$ representative, so that the resulting horizontal $\mathfrak{g}$ -valued 1-form on the glued connection vanishes on $B(x_i, R/2)$ and equals the unmodified residual tangent vector outside $B(x_i, R)$. We may therefore assume a is supported in $\{|x - x_i| \geq R/2\}$, smooth, and harmonic-in-Coulomb in the annulus $\{R/2 \leq |x - x_i| \leq R\}$, with $\|a\|_{L^2}$ controlled by the residual-moduli norm independently of the cutoff. Under this hypothesis,*

$$|\langle a, \hat{v}_\alpha^{(i)} \rangle_{L^2(\mathbb{R}^4)}| \leq C (s_i/R) \|a\|_{L^2} \|\hat{v}_\alpha^{(i)}\|_{L^2}.$$

Proof. By Lemma A.2, $\hat{v}_\alpha^{(i)}$ has pointwise decay $O(s_i^2/r^3)$ at infinity, so the L^1 integral

$$\|\hat{v}_\alpha^{(i)}\|_{L^1(\mathbb{R}^4)} \leq C s_i^2 \int_0^\infty \frac{r^3 dr}{(r + s_i)^3}$$

diverges as $r \rightarrow \infty$ (the integrand approaches the constant $C s_i^2$). A global L^1 - L^∞ pairing is therefore not available; we split into inner and outer regions:

$$|\langle a, \hat{v}^{(i)} \rangle| \leq \underbrace{\|a\|_{L^\infty(B(x_i, R/2))} \|\hat{v}^{(i)}\|_{L^1(B(x_i, R/2))}}_{\text{inner}} + \underbrace{\|a\|_{L^2(\mathbb{R}^4 \setminus B(x_i, R/2))} \|\hat{v}^{(i)}\|_{L^2(\mathbb{R}^4 \setminus B(x_i, R/2))}}_{\text{outer}}.$$

Inner term. Since a is a residual-moduli tangent vector supported by hypothesis at distance $\geq R$ from x_i , a is harmonic (in the linearized ASD sense) and Coulomb on $B(x_i, R)$, hence smooth. The standard four-dimensional Sobolev embedding $H^2(\mathbb{R}^4) \hookrightarrow L^\infty(\mathbb{R}^4)$ with localization to $B(x_i, R/2)$ gives, via the linearized-ASD equation (which is second-order elliptic in Coulomb gauge),

$$\|a\|_{L^\infty(B(x_i, R/2))} \leq C R^{-2} \|a\|_{L^2(B(x_i, R))} \leq C R^{-2} \|a\|_{L^2(\mathbb{R}^4)},$$

where the factor R^{-2} is the scale-invariant Sobolev constant (the embedding $H^2(\mathbb{R}^4) \hookrightarrow L^\infty$ requires two derivatives, each contributing one factor of inverse length; on a ball of radius R this becomes the explicit R^{-2} after rescaling). The inner L^1 norm of $\hat{v}^{(i)}$ on $B(x_i, R/2)$ is

$$\|\hat{v}^{(i)}\|_{L^1(B(x_i, R/2))} \leq C s_i^2 \int_0^{R/2} \frac{r^3 dr}{(r + s_i)^3} \leq C s_i^2 \log(R/s_i) + C s_i R^2.$$

The dominant term is $C s_i R^2$. Combining, inner contribution $\leq C R^{-2} \|a\|_{L^2} \cdot s_i R^2 = C s_i \|a\|_{L^2}$.

We need to convert $\|a\|_{L^2}$ to a bound containing $\|\hat{v}^{(i)}\|_{L^2}$. Note $\|\hat{v}^{(i)}\|_{L^2} = \Theta(1)$ uniformly in s_i (Lemma A.2). Hence the inner term is $\leq C s_i \|a\|_{L^2} \cdot 1 = C s_i \|a\|_{L^2} \|\hat{v}^{(i)}\|_{L^2}$. To extract the claimed s_i^2/R rate, we use that a , being a residual tangent vector, satisfies the additional weighted decay $\|a\|_{L^\infty(B(x_i, R/2))} \leq C R^{-2} \|a\|_{L^2}$ (the scale-invariant Sobolev embedding $H^2(B_R) \hookrightarrow L^\infty(B_R)$ on a ball of radius R has constant $C R^{-2}$ in 4D, two derivatives each contributing R^{-1}) plus the harmonic-function gradient estimate $\|\nabla a\|_{L^\infty(B(x_i, R/2))} \leq C R^{-3} \|a\|_{L^2}$. A first-order Taylor expansion of a around x_i then gives $a(x) - a(x_i) = O(R^{-3} \|a\|_{L^2} \cdot |x - x_i|)$, and the local moment $\int_{B(x_i, R/2)} (x - x_i) \cdot \hat{v}^{(i)} = O(s_i^3)$ (by parity of $\hat{v}$ around x_i) gives an extra factor s_i/R in the inner term, yielding $C s_i^2/R \|a\|_{L^2}$. Combined with $a(x_i)$ -only contribution: the Coulomb-projected $\hat{v}_\alpha^{(i)}$ inherits the centered antisymmetry of $\partial A_\mu^a / \partial s_i^\alpha$ around the bubble center x_i (the BPST profile $A_\mu^a \propto \eta_{\mu\nu}^a (x - x_i)^\nu / D_i$ is odd in $x - x_i$ to leading order, and the gauge correction $d_A \phi^\alpha$ is built from the same

centered source $d_A^* v_\alpha$ so preserves the parity), so each component satisfies $\int_{B(x_i, R/2)} \hat{v}_\alpha^{(i)} d^4x = O(s_i^4)$ — vanishing to leading order in s_i by symmetry, with the higher-order s_i^4 contribution coming from the deviation between Coulomb-projected $\hat{v}$ and the exactly centered profile. Hence the $a(x_i)$ -only piece contributes at most $\|a\|_{L^\infty} \cdot O(s_i^4) = O(s_i^4 R^{-2} \|a\|_{L^2})$, which is subleading to the Taylor remainder. The leading inner contribution is the Taylor-expansion remainder

$$\left| \int_{B(x_i, R/2)} (a(x) - a(x_i)) \cdot \hat{v}^{(i)}(x) d^4x \right| \leq \|\nabla a\|_{L^\infty(B(x_i, R/2))} \cdot \int |x - x_i| |\hat{v}^{(i)}| d^4x \leq \frac{C \|a\|_{L^2}}{R^3} \cdot s_i^3 = C (s_i/R)^3 \|a\|_{L^2}.$$

Outer term. On $\mathbb{R}^4 \setminus B(x_i, R/2)$,

$$\|\hat{v}^{(i)}\|_{L^2(\mathbb{R}^4 \setminus B(x_i, R/2))}^2 \leq C s_i^4 \int_{R/2}^\infty \frac{r^3 dr}{r^6} = O(s_i^4/R^2),$$

so the outer term $\|a\|_{L^2} \cdot O(s_i^2/R)$ dominates the inner $(s_i/R)^3$ contribution. Combining, $|\langle a, \hat{v}^{(i)} \rangle| \leq C (s_i^2/R) \|a\|_{L^2}$. Since $\|\hat{v}^{(i)}\|_{L^2} = \Theta(1)$ this equals $C (s_i^2/R) \|a\|_{L^2} \|\hat{v}^{(i)}\|_{L^2}$, as claimed. $\square$

A.6 Δ_g vs Δ_{prod} bookkeeping

The metric comparison (8) reads $(1 \mp C\epsilon |\log \epsilon|) g_{\text{prod}} \leq g_U \leq (1 \pm C\epsilon |\log \epsilon|) g_{\text{prod}}$ as quadratic forms on tangent vectors. The corresponding comparison of Laplace–Beltrami operators is not direct; it involves the volume Jacobian and a connection contribution. We spell out the bookkeeping.

Let $g = g_{\text{prod}} + h$ with $\|h\|_{g_{\text{prod}}} \leq \epsilon' := C\epsilon |\log \epsilon|$ as a $(0, 2)$ -tensor norm. The dual metric satisfies $g^{-1} = g_{\text{prod}}^{-1} - g_{\text{prod}}^{-1} h g_{\text{prod}}^{-1} + O(\epsilon'^2)$, so $\|g^{-1} - g_{\text{prod}}^{-1}\|_{\text{op}} \leq C\epsilon'$. The volume form $\sqrt{\det g} = \sqrt{\det g_{\text{prod}}} (1 + \frac{1}{2} \text{tr}_{g_{\text{prod}}} h + O(\epsilon'^2))$, with $|\frac{1}{2} \text{tr}_{g_{\text{prod}}} h| \leq C\epsilon'$.

The Laplace–Beltrami operator in coordinates is $\Delta_g f = (\det g)^{-1/2} \partial_i ((\det g)^{1/2} g^{ij} \partial_j f) = g^{ij} \partial_i \partial_j f + (g^{ij} \Gamma_{ij}^k(g)) \partial_k f$ with metric Christoffels. Decomposing,

$$\Delta_g - \Delta_{\text{prod}} = (g^{ij} - g_{\text{prod}}^{ij}) \partial_i \partial_j + \underbrace{(g^{ij} \Gamma_{ij}^k(g) - g_{\text{prod}}^{ij} \Gamma_{ij}^k(g_{\text{prod}}))}_{=: B^k} \partial_k.$$

The leading-coefficient piece $(g^{-1} - g_{\text{prod}}^{-1}) \partial^2$ is bounded by $C\epsilon'$ in operator norm on $H^2 \rightarrow L^2$ (the bound used in the body proof). The first-order remainder $B^k \partial_k$ requires the connection contribution: $\Gamma_{ij}^k(g) - \Gamma_{ij}^k(g_{\text{prod}}) = \frac{1}{2} g^{kl} (\nabla_i^{\text{prod}} h_{jl} + \nabla_j^{\text{prod}} h_{il} - \nabla_l^{\text{prod}} h_{ij})$, hence $|B| \leq C |\nabla h|_{g_{\text{prod}}} \leq C\epsilon' \cdot$ (inverse length scale of the bubble).

In the collar regime, the metric perturbation h and its gradient $\nabla^{\text{prod}} h$ are uniformly bounded in L^∞ on the cusp ends by $C\epsilon'$ (Lemma A.6 at $n = 0, 1$, with the cusp dilation $y_i \partial_{y_i}$ producing a C^0 -bounded unit). The first-order remainder $B^k \partial_k$ in the operator decomposition of $\Delta_g - \Delta_{\text{prod}}$ thus has coefficients bounded by $C\epsilon'$ in L^∞ , and Cauchy–Schwarz combined with the standard H^1 elliptic bound $\|\nabla \phi\|_{L^2}^2 \leq \langle -\Delta_{\text{prod}} \phi, \phi \rangle$ gives directly

$$|\langle B^k \partial_k \phi, \phi \rangle| \leq C\epsilon' \|\nabla \phi\|_{L^2} \|\phi\|_{L^2} \leq C\epsilon' \langle (-\Delta_{\text{prod}} + 1) \phi, \phi \rangle.$$

This unweighted bookkeeping avoids the need for a Hardy-type weight on the hyperbolic-cusp coordinate.

Combining: $|\langle (\Delta_g - \Delta_{\text{prod}}) \phi, \phi \rangle| \leq C\epsilon' \langle -\Delta_{\text{prod}} \phi, \phi \rangle$, i.e., a quadratic-form comparison with the same multiplicative rate $\epsilon' = O(\epsilon |\log \epsilon|)$ as the metric comparison. This is the input to the min-max argument in the proof of Theorem 4.3. The volume-form Jacobian $\sqrt{\det g} / \sqrt{\det g_{\text{prod}}} = 1 + O(\epsilon')$

shifts the $L^2(d\text{vol}_g)$ inner product by a multiplicative $(1 + O(\epsilon'))$ relative to $L^2(d\text{vol}_{\text{prod}})$; this affects only the absolute normalization of the quadratic form, not the spectral location, and is absorbed into the implicit constant C in the body proof.

This completes the proof of Lemma 4.2(iv)–(v) and the associated Δ_g -vs- Δ_{prod} comparison used in Theorem 4.3.

A.7 Iterated dilation derivatives and the second-commutator bound

For Theorem 4.8 we need a bound on iterated commutators of the cusp-dilation conjugate operator A_j against the metric Laplacian. The single-commutator bound is essentially §A.6; the new content is the second-commutator bound and, more generally, control of $(y_i \partial_{y_i})^n h$ for the perturbation tensor.

Recall (Lemma A.2) that on each bubble factor, the cusp coordinate y_i in the upper-half-space model of $\mathbb{H}_r^5$ is related to the BPST scale parameter s_i by $y_i \sim r^2/s_i$ (the cusp coordinate is the inverted scale). The cusp-dilation $y_i \partial_{y_i}$ acts on functions of s_i via $y_i \partial_{y_i} = -s_i \partial_{s_i}$ (the sign is immaterial below; only the magnitude of iterated derivatives enters the bound).

Lemma A.6 (Iterated dilation derivatives of the perturbation tensor). *Let $h := g_{U_\epsilon^{(j)}} - g_{\text{prod}}$ be the metric perturbation on the codimension- j Uhlenbeck collar, expressed in the gauge-fixed bubble-background coordinate system (§A.4–§A.5). For every integer $n \geq 0$ there is a constant $C_n = C_n(j, r)$ such that, pointwise on the cusp ends of $U_\epsilon^{(j)}$ (i.e., where $\max_i s_i \leq \epsilon$, equivalently $\min_i y_i \geq Y_0(\epsilon)$),*

$$\left\| \left(\sum_{i=1}^j y_i \partial_{y_i} \right)^n h \right\|_{g_{\text{prod}}} \leq C_n \epsilon |\log \epsilon| (1 + |\log \epsilon|)^n. \quad (14)$$

The same bound, with the same rate $\epsilon |\log \epsilon| (1 + |\log \epsilon|)^n$, holds for $(\sum_i y_i \partial_{y_i})^n \nabla^{\text{prod}} h$, with one extra inverse-length factor of $1/r_{\text{cusp}}$ on the right-hand side (so that the Hardy inequality of §A.6 absorbs the gradient).

Proof. By Lemma 4.2(iv)–(v), the cross-block entries of h in the gauge-fixed basis are linear combinations of the inner products $\langle \hat{v}_\alpha^{(i)}, \hat{v}_\beta^{(j')} \rangle_{L^2(\mathbb{R}^4)}$ and $\langle a, \hat{v}_\alpha^{(i)} \rangle_{L^2(\mathbb{R}^4)}$, with explicit dependence on the bubble parameters (s_i, x_i) through the BPST profiles $A_i(x; x_i, s_i)$. By the scale-conformal invariance of the BPST family $A(x; x_0, s) = s^{-1} A_*((x - x_0)/s)$ for a fixed profile A_* on $\mathbb{R}^4$ (canonical centered unit-scale BPST), the gauge-projected tangent vectors transform homogeneously:

$$\hat{v}_s^{(i)}(x; x_i, s_i) = \hat{V}_s((x - x_i)/s_i), \quad \hat{v}_{x,\rho}^{(i)}(x; x_i, s_i) = s_i^{-1} \hat{V}_{x,\rho}((x - x_i)/s_i), \quad (15)$$

for fixed model fields $\hat{V}_s, \hat{V}_{x,\rho}$ on $\mathbb{R}^4$ depending only on the unit-scale BPST connection. Consequently, in the integration variable $\xi := (x - x_i)/s_i$,

$$\langle \hat{v}_\alpha^{(i)}, \hat{v}_\beta^{(j')} \rangle_{L^2(\mathbb{R}^4)} = s_i^{4-w_\alpha} s_{j'}^{-w_\beta} \int_{\mathbb{R}^4} \hat{V}_\alpha(\xi) \cdot \hat{V}_\beta((s_i \xi + x_i - x_{j'})/s_{j'}) d^4 \xi,$$

where $w_s = 0$, $w_x = 1$ are the weights from (15).

Effect of the dilation $s_i \partial_{s_i}$ on cross-block entries. Apply $s_i \partial_{s_i}$ to the right-hand side. The prefactor $s_i^{4-w_\alpha}$ contributes a multiplicative constant $(4-w_\alpha)$. The argument of the second integrand $\hat{V}_\beta$ contains $s_i \xi/s_{j'}$, and $s_i \partial_{s_i}$ acting on $\hat{V}_\beta((s_i \xi + x_i - x_{j'})/s_{j'})$ gives $(s_i \xi/s_{j'}) \cdot (\nabla \hat{V}_\beta)(\dots) = (\nabla \hat{V}_\beta \cdot \xi) \cdot (s_i/s_{j'})$. Writing this back as the integral of $\hat{V}_\alpha \cdot (\nabla \hat{V}_\beta \cdot \xi)$ against the same Jacobian, we obtain a new bilinear form on the model BPST fields that has the *same* field-strength decay

rate $|\hat{V}_\alpha \nabla \hat{V}_\beta \cdot \xi| = O(|\xi|/(1+|\xi|)^7) = O(1/(1+|\xi|)^6)$ as the original integrand (Lemma A.2: $|\hat{V}_\alpha| = O(1/(1+|\xi|)^3)$, $|\nabla \hat{V}_\beta| = O(1/(1+|\xi|)^4)$). The model-space integral remains absolutely convergent. Iterating: $(s_i \partial_{s_i})^n$ produces a finite linear combination of integrals of the schematic form $\int \hat{V}_\alpha \cdot D^{(n)} \hat{V}_\beta \cdot d^4 \xi$, where $D^{(n)}$ is a polynomial-in- ξ differential operator of order $\leq n$, each of which has the same model-space convergence by the field-strength decay applied $n+1$ times. The result is that iterated $s_i \partial_{s_i}$ does *not* degrade the cross-block decay rate from Lemma 4.2(iv):

$$|(s_i \partial_{s_i})^n \langle \hat{v}_\alpha^{(i)}, \hat{v}_\beta^{(j')} \rangle_{L^2}| \leq C_n \frac{s_i s_{j'}}{R^2} (1 + |\log(\min(s_i, s_{j'})/R)|)^{1+m_n}, \quad (16)$$

where $m_n \leq n$ tracks the maximum number of additional logarithm factors introduced by $s_i \partial_{s_i}$ acting on the $\log(s/R)$ borderline factor of the position-position case (Lemma 4.2(iv)). Each application of $s_i \partial_{s_i}$ to $\log(s_i/R)$ produces a constant 1 (not an extra inverse $1/s_i$), since $s_i \partial_{s_i} \log(s_i/R) = 1$; subsequent applications annihilate the constant. Hence at most one extra log factor is produced per application of $s_i \partial_{s_i}$, giving the bound $(1 + |\log \epsilon|)^{n+1}$ on the right.

Bubble-background entries. By the same argument applied to Proposition A.5, $(s_i \partial_{s_i})^n \langle a, \hat{v}_\alpha^{(i)} \rangle = O(s_i^2/R) \cdot (1 + |\log \epsilon|)^n \cdot \|a\|_{L^2} \|\hat{v}\|_{L^2}$. The decay rate s_i^2/R is preserved by iterated $s \partial_s$ for the same reason as above.

Conversion to the cusp dilation. On the cusp end of the i -th bubble factor, $y_i \partial_{y_i} = -s_i \partial_{s_i}$ (the cusp coordinate is $y_i \sim r^2/s_i$, so $\log y_i = 2 \log r - \log s_i$, giving $\partial/\partial \log y_i = -\partial/\partial \log s_i$). Iterated $(y_i \partial_{y_i})^n$ thus produces $(-1)^n (s_i \partial_{s_i})^n$ on the cross-block entries. The sum over i in A_j yields a multinomial expansion of $(\sum_i y_i \partial_{y_i})^n$ into products of $(y_{i_1} \partial_{y_{i_1}}) \cdots (y_{i_n} \partial_{y_{i_n}})$ acting on a single cross-block entry $\langle \hat{v}^{(i)}, \hat{v}^{(j')} \rangle$, which is nonzero only when $\{i_1, \dots, i_n\} \subseteq \{i, j'\}$ by the scale-conformal invariance. Each such product is controlled by (16) with the same overall rate.

Combining over the $j \times j + 2j(k-j)$ entries of the bubble-bubble and bubble-background cross-blocks of h , with $\max_i s_i \leq \epsilon$:

$$\left\| \left(\sum_i y_i \partial_{y_i} \right)^n h \right\|_{g_{\text{prod}}} \leq C_n (\epsilon^2/R^2 + \epsilon^2/R) (1 + |\log \epsilon|)^{n+1} \leq C'_n \epsilon |\log \epsilon| (1 + |\log \epsilon|)^n \quad \text{for } \epsilon \leq \epsilon_0(c, r) := \min\{e^{-1}, \dots\}$$

absorbing one factor of ϵ and using $R = \Theta(1)$ uniformly on the collar (the inter-bubble separation is bounded below by the diameter of S_r^4 minus the bubble scales, hence $\geq r - O(\epsilon)$). This is (14).

The gradient version follows by differentiating once more and noting that ∇^{prod} contributes an extra inverse-length factor $1/r_{\text{cusp}}$ in the hyperbolic metric. $\square$

Lemma A.7 (Iterated-commutator bound). *For each $n \geq 1$, the iterated commutator*

$$\text{ad}_{iA_j}^{(n)}(-\Delta_{U_\epsilon^{(j)}}) - \text{ad}_{iA_j}^{(n)}(-\Delta_{\text{prod}})$$

extends from $C_c^\infty(U_\epsilon^{(j)})$ to a quadratic form satisfying

$$|\langle (\text{ad}_{iA_j}^{(n)}(-\Delta_{U_\epsilon^{(j)}}) - \text{ad}_{iA_j}^{(n)}(-\Delta_{\text{prod}})) \phi, \phi \rangle| \leq C_n \epsilon (1 + |\log \epsilon|)^{n+1} \langle (-\Delta_{\text{prod}} + 1) \phi, \phi \rangle \quad (17)$$

for $\phi \in C_c^\infty(U_\epsilon^{(j)})$. In particular for $n = 2$ the right-hand side is $O(\epsilon |\log \epsilon|^3)$, which is the $C^{1,1}$ (in fact C^2) regularity input for Theorem 4.8(ii).

Proof. From §A.6,

$$-\Delta_{U_\epsilon^{(j)}} + \Delta_{\text{prod}} = (g^{-1} - g_{\text{prod}}^{-1})_{ij} \partial^i \partial^j + B^k \partial_k =: L_h,$$

with L_h a second-order operator whose coefficients are smooth functions of h and $\nabla^{\text{prod}}h$, polynomial in the coordinates of h (and g_{prod}^{-1} , which is bounded uniformly on the cusp). The dilation generator A_j commutes with $-\Delta_{\text{prod}}$ up to the explicit identity $[-\Delta_{\text{prod}}, iA_j] = 2(-\Delta_{\text{prod}} - 4j/r^2) + K_{\text{cpt}}$ in the conjugated radial Schrödinger frame (10)–(11) of §4.4 (where K_{cpt} collects the cutoff χ error, the spherical-harmonic Casimir contribution from sectors $\ell \geq 1$, and the half-density conjugation residual, all relatively compact perturbations; cf. Froese–Hislop [21]).

Iterated commutators with A_j act on L_h by Lie differentiation along the dilation flow plus polynomial-coefficient lower-order corrections:

$$[L_h, iA_j] = L_{\mathcal{L}_{A_j}h} + [L_h, iA_j]_{\text{l.o.t.}},$$

where $\mathcal{L}_{A_j}$ is the Lie derivative along the dilation vector field $A_j = \sum_i y_i \partial_{y_i}$. The lower-order term $[L_h, iA_j]_{\text{l.o.t.}}$ is a first-order differential operator with coefficients of the same form as $h, \nabla^{\text{prod}}h$ (paired with the bounded coefficients of A_j), of the same operator-norm size as L_h itself. Iterating n times produces $L_{\mathcal{L}_{A_j}^n h}$ plus lower-order analogues with coefficients in $(\mathcal{L}_{A_j}^k h)_{0 \leq k \leq n}$.

By Lemma A.6, $\|\mathcal{L}_{A_j}^k h\|_{g_{\text{prod}}} \leq C_k \epsilon (1 + |\log \epsilon|)^{k+1}$ pointwise on the cusp ends. Substituting into the operator-norm decomposition $L_h = (g^{-1} - g_{\text{prod}}^{-1})\partial^2 + B^k \partial_k$, applying the same Hardy-inequality manoeuvre of §A.6 to absorb the first-order term $B^k \partial_k$ against $-\Delta_{\text{prod}} + 1$, and bounding the ∂^2 term by $\langle (-\Delta_{\text{prod}} + 1)\phi, \phi \rangle$ via standard H^2 -elliptic regularity, we obtain

$$| \langle L_{\mathcal{L}_{A_j}^k h} \phi, \phi \rangle | \leq C \|\mathcal{L}_{A_j}^k h\|_{g_{\text{prod}}} \langle (-\Delta_{\text{prod}} + 1)\phi, \phi \rangle \leq C_k \epsilon (1 + |\log \epsilon|)^{k+1} \langle (-\Delta_{\text{prod}} + 1)\phi, \phi \rangle.$$

Summing over $k \leq n$ (with multinomial constants absorbed into C_n) gives (17).

For $n = 1$ this recovers the single-commutator bound used in the proof of Theorem 4.3 (with the simpler $\epsilon |\log \epsilon|$ rate, since only $k = 0, 1$ enter). For $n = 2$ the bound is $O(\epsilon |\log \epsilon|^3)$, which tends to 0 as $\epsilon \rightarrow 0$, so the iterated commutator is uniformly bounded relative to $-\Delta_{\text{prod}} + 1$ and in particular relative to $-\Delta_{U_\epsilon^{(j)}} + 1$ (since the two are mutually bounded by §A.6). This is precisely the $C^{1,1}(-\Delta_{U_\epsilon^{(j)}})$ -regularity hypothesis of Sahbani [17, Hyp. H1–H3]; in fact, since the bound is $O(\epsilon |\log \epsilon|^3) \rightarrow 0$, we obtain the stronger $C^2(-\Delta_{U_\epsilon^{(j)}})$ regularity. $\square$

Remark A.8 (Comparison with the bubble-background derivative). The iterated bound (14) is uniform: at each order n the right-hand side gains one logarithmic factor but does not gain inverse-bubble-scale factors $1/s_i$ that would conflict with the ϵ decay. The crucial mechanism is that $s_i \partial_{s_i}$ acting on the BPST scale-homogeneous profile produces tangent vectors of the same scale class (Lie derivative along the conformal scale-vector field), not gradient-type terms losing decay. This is what fails in the analogous attempt to use position-dilation $(x - x_i) \cdot \partial_x$ as a conjugate operator: that would lose one inverse power of the bubble scale per derivative, breaking the iterated bound at $n = 2$. The cusp coordinate y_i (inverted scale) is the geometrically correct choice.

This completes the second-commutator bound needed for Theorem 4.8.

References

- [1] I.M. Singer, *The geometry of the orbit space for nonabelian gauge theories*, Phys. Scripta **24** (1981) 817–820.
- [2] D. Groisser, T. Parker, *The Riemannian geometry of the Yang–Mills moduli space*, Comm. Math. Phys. **112** (1987) 663–689.

- [3] D. Groisser, T. Parker, *The geometry of the Yang–Mills moduli space for definite manifolds*, J. Differential Geom. **29** (1989) 499–544.
- [4] L. Habermann, *The L^2 -metric on the moduli space of $SU(2)$ -instantons with instanton number 1 over the Euclidean 4-space*, Ann. Glob. Anal. Geom. **11** (1993) 311–322.
- [5] H. Doi, Y. Matsumoto, T. Matsumoto, *An explicit formula of the metric on the moduli space of BPST-instantons over S^4* , A Fête of Topology, Academic Press (1987).
- [6] H.P. McKean, *An upper bound to the spectrum of Δ on a manifold of negative curvature*, J. Differential Geom. **4** (1970) 359–366.
- [7] C.H. Taubes, *A framework for Morse theory for the Yang–Mills functional*, Invent. Math. **94** (1988) 327–402.
- [8] R. Bartnik, *The mass of an asymptotically flat manifold*, Comm. Pure Appl. Math. **39** (1986) 661–693.
- [9] M.F. Atiyah, N.J. Hitchin, I.M. Singer, *Self-duality in four-dimensional Riemannian geometry*, Proc. Roy. Soc. London A **362** (1978) 425–461.
- [10] V. Bonfili, Sage symbolic and numerical verification of the $SU(N)$ extension of Lemma 4.1: `verification/lemma_ct_SUN.sage`. Confirms the $SU(N)$ common-subgroup cross-term reproduces the $SU(2)$ closed form to $\sim 10^{-14}$ relative precision at $N = 2, 3, 4$, and computes the framing-overlap scalar $o(U) \in [-1/3, 1]$ on random samples.
- [11] R.B. Lockhart, R.C. McOwen, *Elliptic differential operators on noncompact manifolds*, Ann. Scuola Norm. Sup. Pisa Cl. Sci. **12** (1985) 409–447.
- [12] S.K. Donaldson, P.B. Kronheimer, *The Geometry of Four-Manifolds*, Oxford Univ. Press (1990).
- [13] V. Bonfili, Schwinger-parametrized closed form for the BPST scale-derivative cross-term, with uniform-box Monte Carlo cross-check: `verification/lemma_ct_montecarlo.py`.
- [14] V. Bonfili, machine-precision Rust verification of Lemma 4.1: `verification/lemma_ct_rust/`. Self-contained adaptive Gauss–Kronrod 15/7 in Rust; reduces the 4-d integral to 2-d via axial $O(3)$ symmetry, then compares against the 1-d Schwinger ground truth. Relative error ≤ 2.5 ulp ($\approx 5.5 \times 10^{-16}$) across a 106-point grid spanning $(s_1, s_2, R) \in [0.05, 2.0]^2 \times [1, 50]$ plus extreme asymptotic regimes; runtime < 5 s on a Mac mini.
- [15] V. Bonfili, Sage symbolic computation of the next-order Schwinger expansion for Lemma 4.1: `verification/lemma_ct_higher_order.sage`. Verifies $F(u, 0) = 1/(1 + u^2)$ exactly, the symmetric series $F(\epsilon, \epsilon) = 1 - 2\epsilon^2 + 8(1 + \log \epsilon)\epsilon^4 + O(\epsilon^6 \log \epsilon)$, and the identification of the $\log(s/R)$ remainder at the $u^2 v^2$ cross-term.
- [16] V. Bonfili, symbolic and high-precision verification of the elementary and hyperbolic-distance closed forms (3)–(4): `verification/lemma_ct_elementary.py`. Verifies the pointwise integrand identity and the numerator identity $N = \Sigma D$ symbolically (exact); the closed form against 50-digit adaptive quadrature of the integral in (1) on a 60-point grid (max. rel. error $< 10^{-45}$); the full inner product against the Rust 4-D grid values of [14] (machine precision);

- the limits $G(0^+) = 2/3$ and $G(d) \sim 2e^{-d}$; and the symmetric expansion coefficient $8(1 + \log \epsilon)$ of [15] directly from the closed form.
- [17] J. Sahbani, *The conjugate operator method for locally regular Hamiltonians*, J. Operator Theory **38** (1997) 297–322.
 - [18] E.M. Stein, *Singular Integrals and Differentiability Properties of Functions*, Princeton Math. Series **30**, Princeton Univ. Press (1970). Ch. V, Thm. 1 and Cor. 2, pp. 119–121, for the weighted Riesz-composition / Hardy–Littlewood–Sobolev convolution estimate used in Lemma A.3 (the composition estimate is Cor. 2).
 - [19] M. Reed, B. Simon, *Methods of Modern Mathematical Physics II: Fourier Analysis, Self-Adjointness*, Academic Press (1975). Theorem X.39 (Nelson’s analytic vector theorem) for essential self-adjointness of the cutoff dilation generator A_j on C_c^∞ .
 - [20] W.O. Amrein, A. Boutet de Monvel, V. Georgescu, *C_0 -Groups, Commutator Methods and Spectral Theory of N -Body Hamiltonians*, Progress in Mathematics **135**, Birkhäuser (1996).
 - [21] R. Froese, P. Hislop, *Spectral analysis of second-order elliptic operators on noncompact manifolds*, Duke Math. J. **58** (1989) 103–129.
 - [22] V. Bonfioli, Sage symbolic verification of the $\mathbb{H}_7^5$ Mourre commutator identity and the iterated-commutator bound for the perturbation tensor: `verification/mourre_estimate_cusp.sage` and `verification/mourre_iterated_commutator.sage`.
 - [23] V. Bonfioli, Sage derivation and numerical verification of the bubble-bubble cross-block closed forms for Lemma 4.2 (scale-position and the IR-structure of position-position): `verification/lemma_od_position_terms.sage`. Computes the exact Schwinger closed form (with denominator $X(t)^2$ from the squared propagators after σ -integration), confirms the leading asymptotic (6) to relative error $< 10^{-3}$ across (R, s_1, s_2) ranging over four orders of magnitude, identifies the logarithmic infrared divergence of bare position-position cross-terms, and documents the gauge-projection argument leading to (7).